

國立成功大學
統計學研究所

統計套裝軟體講義
R 軟體操作

授課老師：馬瀾嘉

中華民國 105 年 2 月

R 的官方網站 <http://www.r-project.org/>

>Download, Packages

CRAN

>Windows

>base

>Download R 3.0.3 for Windows

R version 3.0.3 (2014-03-06) -- "Warm Puppy"

Copyright (C) 2014 The R Foundation for Statistical Computing

Platform: i386-w64-mingw32/i386 (32-bit)

R 是免費軟體，不提供任何擔保。

在某些條件下您可以將其自由散布。

用 'license()' 或 'licence()' 來獲得散布的詳細條件。

R 是個合作計劃，有許多人為之做出了貢獻。

用 'contributors()' 來看詳細的情況並且

用 'citation()' 會告訴您如何在出版品中正確地參照 R 或 R 套件。

用 'demo()' 來看一些示範程式，用 'help()' 來檢視線上輔助檔案，或

用 'help.start()' 透過 HTML 瀏覽器來看輔助檔案。

用 'q()' 離開 R。

輸入資料(1,2,3)

> x=c(1,2,3)

> x

[1] 1 2 3

簡單計算

> (2+5)*6/(3-1)

[1] 21

資料(1,2,3)，每個元素取倒數、取 Log

> 1/x

[1] 1.0000000 0.5000000 0.3333333

> log(x)

[1] 0.0000000 0.6931472 1.0986123

> y=c(4,3,2)

> x+y

[1] 5 5 5

> (3*sqrt(-1+0i)+2)/(sqrt(-2+0i))

[1] 2.12132-1.414214i

```

> a=seq(1,100,by=1)
> a
 [1]  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17
18
[19] 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
36
[37] 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53
54
[55] 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71
72
[73] 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89
90
[91] 91 92 93 94 95 96 97 98 99 100
> cumsum(a)
 [1]  1  3  6 10 15 21 28 36 45 55 66 78 91 105
[15] 120 136 153 171 190 210 231 253 276 300 325 351 378 406
[29] 435 465 496 528 561 595 630 666 703 741 780 820 861 903
[43] 946 990 1035 1081 1128 1176 1225 1275 1326 1378 1431 1485 1540 1596
[57] 1653 1711 1770 1830 1891 1953 2016 2080 2145 2211 2278 2346 2415 2485
[71] 2556 2628 2701 2775 2850 2926 3003 3081 3160 3240 3321 3403 3486 3570
[85] 3655 3741 3828 3916 4005 4095 4186 4278 4371 4465 4560 4656 4753 4851
[99] 4950 5050
> as=a^2
> cumsum(as)
 [1]  1  5 14 30 55 91 140 204 285 385
[11] 506 650 819 1015 1240 1496 1785 2109 2470 2870
[21] 3311 3795 4324 4900 5525 6201 6930 7714 8555 9455
[31] 10416 11440 12529 13685 14910 16206 17575 19019 20540 22140
[41] 23821 25585 27434 29370 31395 33511 35720 38024 40425 42925
[51] 45526 48230 51039 53955 56980 60116 63365 66729 70210 73810
[61] 77531 81375 85344 89440 93665 98021 102510 107134 111895 116795
[71] 121836 127020 132349 137825 143450 149226 155155 161239 167480 173880
[81] 180441 187165 194054 201110 208335 215731 223300 231044 238965 247065
[91] 255346 263810 272459 281295 290320 299536 308945 318549 328350 338350
> a1=seq(length=50,from=5,by=3)
> a1
 [1]  5  8 11 14 17 20 23 26 29 32 35 38 41 44 47 50 53
56
[19] 59 62 65 68 71 74 77 80 83 86 89 92 95 98 101 104 107 110
[37] 113 116 119 122 125 128 131 134 137 140 143 146 149 152

```

```
## 輸入成績
```

```
> x1=read.table("d:/score.txt")
```

```
> x1
```

```
      V1
```

```
1  86
```

```
2  75
```

```
3  75
```

```
4  94
```

```
5  37
```

```
6  86
```

```
7  90
```

```
8  78
```

```
9  75
```

```
10 72
```

```
11 75
```

```
12 70
```

```
13 64
```

```
14 65
```

```
15 70
```

```
16 63
```

```
17 73
```

```
18 88
```

```
19 56
```

```
20 74
```

```
21 75
```

```
22 73
```

```
23 89
```

```
24 62
```

```
25 72
```

```
> dim(x1)
```

```
[1] 25  1
```

```
> x=scan(file="d:/score.txt")
```

```
Read 25 items
```

```
> x
```

```
[1] 86 75 75 94 37 86 90 78 75 72 75 70 64 65 70 63 73 88 56 74 75 73 89 62 72
```

```
> great= x>85
```

```
> great
```

```
[1] TRUE FALSE FALSE TRUE FALSE TRUE TRUE FALSE FALSE FALSE FALSE
```

```

FALSE
[13] FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE TRUE
FALSE FALSE
> sum(great)
[1] 6
> rrr=(x<70)&(x>60)
> rrr
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
FALSE
[13] TRUE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE FALSE
TRUE FALSE
> sum(rrr)
[1] 4
> sum((x>80)|(x<30))
[1] 6
> x[1:20]
[1] 86 75 75 94 37 86 90 78 75 72 75 70 64 65 70 63 73 88 56 74

```

##做成矩陣

```

> xarray=array(x,dim=c(5,5))
> xarray
      [,1] [,2] [,3] [,4] [,5]
[1,]   86   86   75   63   75
[2,]   75   90   70   73   73
[3,]   75   78   64   88   89
[4,]   94   75   65   56   62
[5,]   37   72   70   74   72
> xarray=matrix(x,5,5)
> xarray
      [,1] [,2] [,3] [,4] [,5]
[1,]   86   86   75   63   75
[2,]   75   90   70   73   73
[3,]   75   78   64   88   89
[4,]   94   75   65   56   62
[5,]   37   72   70   74   72
> z=matrix(0,5,6)
> z
      [,1] [,2] [,3] [,4] [,5] [,6]
[1,]    0    0    0    0    0    0
[2,]    0    0    0    0    0    0

```

```
[3,] 0 0 0 0 0 0
[4,] 0 0 0 0 0 0
[5,] 0 0 0 0 0 0
```

##從 1~9 以取後放回抽樣，抽出 15 筆資料

```
> x=sample(1:9,15,replace=TRUE)
```

```
> A=matrix(x,3,5)
```

```
> A
```

```
      [,1] [,2] [,3] [,4] [,5]
[1,]    6    7    6    8    1
[2,]    6    7    3    2    5
[3,]    9    4    5    9    3
```

```
> y=sample(1:12,10,replace=TRUE)
```

```
> B=matrix(y,5,2)
```

```
> B
```

```
      [,1] [,2]
[1,]   12    5
[2,]    4    4
[3,]    6    7
[4,]    3   10
[5,]    5    7
```

##矩陣相乘

```
> A%%B
```

```
      [,1] [,2]
[1,]  165  187
[2,]  149  134
[3,]  196  207
```

```
> A*A
```

```
      [,1] [,2] [,3] [,4] [,5]
[1,]   36   49   36   64    1
[2,]   36   49    9    4   25
[3,]   81   16   25   81    9
```

```
> A%%A
```

錯誤在 A %% A：非調和引數

##解聯立方程式

```
> A=matrix(c(2,1,3,-5),2,2)
```

```
> y=matrix(c(-7,9),2,1)
```

```
> x=solve(A,y)
```

```

> x
      [,1]
[1,] -0.6153846
[2,] -1.9230769
> solve(A)
      [,1]      [,2]
[1,] 0.38461538 0.2307692
[2,] 0.07692308 -0.1538462
> solve(A)%*%y
      [,1]
[1,] -0.6153846
[2,] -1.9230769

```

##生成標準常態分配資料

```

> x=rnorm(10)
> x
[1] 0.47481003 0.24329626 0.80276658 -0.31315821 0.23896421 1.09570599
[7] -0.97353837 0.76722181 -0.06063246 0.52453604
> ifelse(x>=0,x,-x)
[1] 0.47481003 0.24329626 0.80276658 0.31315821 0.23896421 1.09570599
[7] 0.97353837 0.76722181 0.06063246 0.52453604
> abs(x)
[1] 0.47481003 0.24329626 0.80276658 0.31315821 0.23896421 1.09570599
[7] 0.97353837 0.76722181 0.06063246 0.52453604

```

##計算 100 的階乘 100!

```

> fac=function(x){prod=1
+ for (i in 1:x) {prod=prod*i}
+ return(prod)
+ }
> fac(100)
[1] 9.332622e+157
> pi
[1] 3.141593
> exp(1)
[1] 2.718282
> n=100
> ss=sqrt(2*pi)*n^(n+0.5)*exp(-n)
> ss
[1] 9.324848e+157

```

##畫莖葉圖

```
> score=scan(file="d:/score.txt")
```

Read 50 items

```
> stem(score)
```

The decimal point is 1 digit(s) to the right of the |

3 | 7

4 | 9

5 | 6

6 | 2334555679

7 | 00223333455555555568889

8 | 0112466789

9 | 0146

##計算平均數和變異數

```
> mean(score)
```

[1] 74.4

```
> var(score)
```

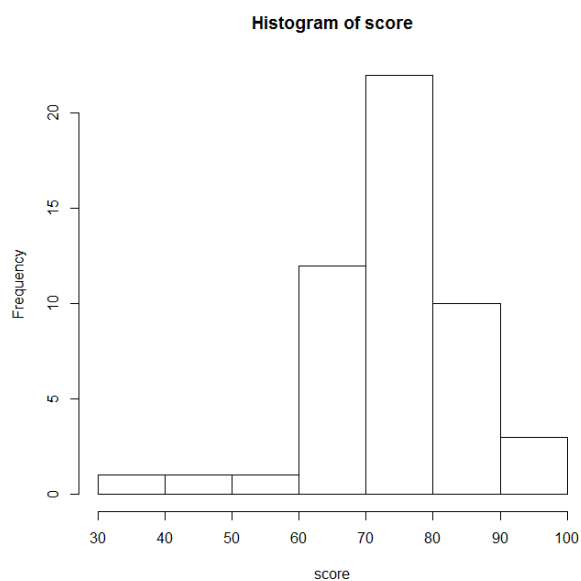
[1] 120.6531

```
> summary(score)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
37.00	69.25	75.00	74.40	80.75	96.00

##畫直方圖

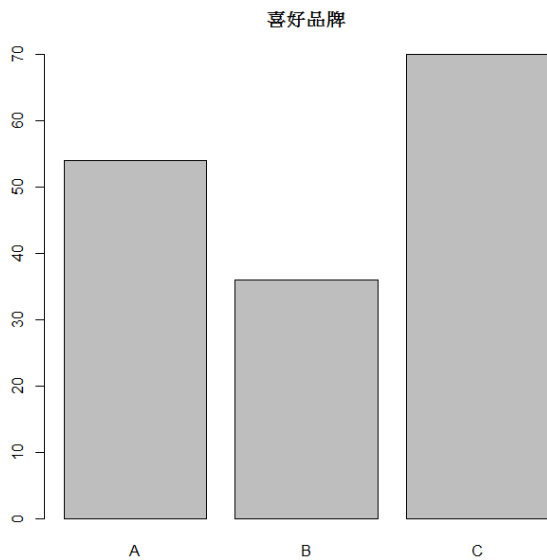
```
> hist(score)
```



```
> stripchart(score,method="stack",xlab="score",pch=1,offset=1,cex=1)
```

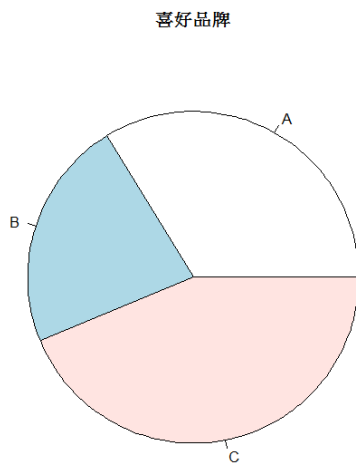

##畫長條圖

```
> brand=c(54,36,70)
> names(brand)=c("A","B","C")
> barplot(brand, main="喜好品牌")
```



##畫圓餅圖

```
> pie(brand,main="喜好品牌")
```



##生成常態分配資料 N(mean, variance)

```
> y1=rnorm(100, mean=165, sd=6)
> y2=rnorm(100, mean=170, sd=6)
> summary(y1)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
152.5	162.5	165.0	165.0	167.5	177.5

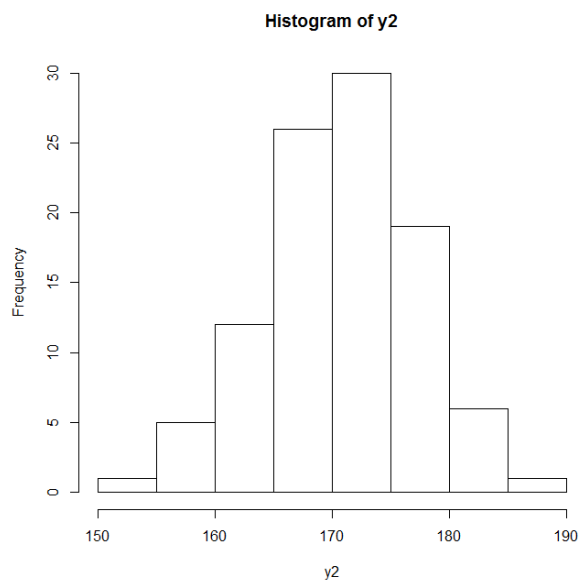
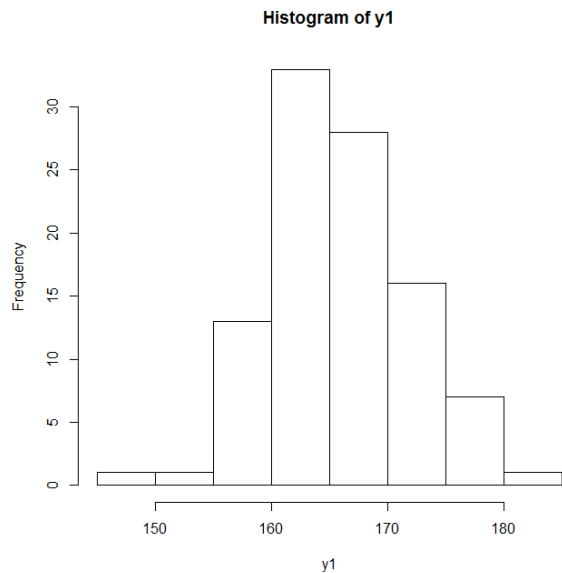
149.4 161.9 165.3 165.7 169.6 182.6

```
> summary(y2)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
152.0	166.2	171.0	170.8	175.1	188.7

```
> hist(y1)
```

```
> hist(y2)
```



##從 101~200 抽出 10 個號碼

```
> x=sample(101:200,10)
```

```
> x
```

```
[1] 101 194 118 189 166 124 137 150 120 142
```

```
> tt=c(rep(1,12),rep(0,18))
```

```
> tt
```

```
[1] 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
> xx=sample(tt,5000,replace=TRUE)
```

```
> mean(xx)
```

```
[1] 0.3992
```

```
##計算單尾檢定所需樣本數
```

```
> samp=function(a,b,d,s){n=(qnorm(1-a)+qnorm(1-b))^2/(d/s)^2;n}
```

```
> samp(0.05,0.2,10,30)
```

```
[1] 55.64302
```

```
##計算檢定兩組比例是否相等所需樣本數
```

```
> samp.ratio=function(a,b,pt,pc)
```

```
+ {n=(qnorm(1-a/2)+qnorm(1-b))^2*(pt*(1-pc)+pc*(1-pt))/(pt-pc)^2;n};
```

```
> print(c("the required sample size for ratio data is",
```

```
+ samp.ratio(0.05,0.2,0.35,0.25)))
```

```
[1] "the required sample size for ratio data is"
```

```
[2] "333.577388709836"
```

```
##計算中位數
```

```
> x=c(21,20,15,14,16,15,19,16,18,19)
```

```
> sort(x)
```

```
[1] 14 15 15 16 16 18 19 19 20 21
```

```
> length(x)
```

```
[1] 10
```

```
> median(x)
```

```
[1] 17
```

```
##計算卜瓦松分配 POI(2.74) 的機率函數和累積分配函數
```

```
> y=c(0,1,2,3,4)
```

```
> py=dpois(y,2.74)
```

```
> py
```

```
[1] 0.06457035 0.17692275 0.24238417 0.22137754 0.15164362
```

```
> ppois(5,2.74)
```

```
[1] 0.9399991
```

```
##生成指數分配之隨機變數、計算其累加值並作圖
```

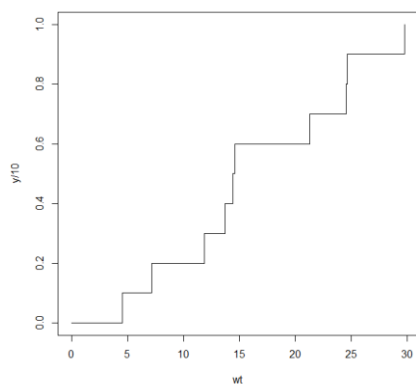
```
> n=10
```

```
> inter=rexp(n,rate=0.3)
```

```
> wt=c(0,cumsum(inter))
```

```
> y=seq(from=0,to=n,by=1)
```

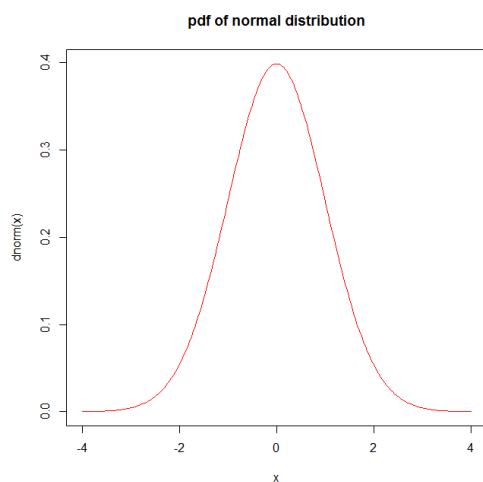
```
> plot(wt,y/10,type="s")
```



##畫出標準常態分配 $N(0,1)$ 的機率密度函數圖形

```
> x=seq(-4,4,length=200)
```

```
> plot(x,dnorm(x),type='l',col='red',main='pdf of normal distribution')
```



##計算常態分配的 pdf, cdf 和 percentile

```
> pnorm(1,4)
```

```
[1] 0.001349898
```

```
> dnorm(1.5)
```

```
[1] 0.1295176
```

```
> qnorm(0.975)
```

```
[1] 1.959964
```

##從 $U(0,1)$ 生成 y ，如果 $y \geq 0.1$ 則從 $N(0,1)$ 生成 z ，否則從 $N(0,9)$ 生成 z

```
> y=runif(1)
```

```
> if (y>=0.1){z=rnorm(1)}else
```

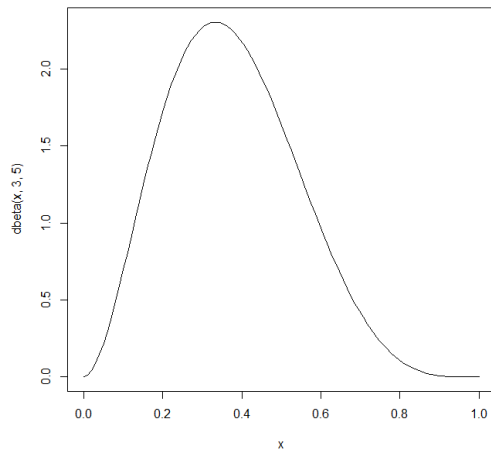
```
+ {z=rnorm(1,mean=0,sd=3)};
```

```
> print(c(y,z))
```

```
[1] 0.6242757 -0.5898405
```

##畫出 Beta 分配的機率密度函數圖形

```
> x <- seq(0,1,length=100)
> plot(x,dbeta(x,3,5),type='l')
```



##Harrell-Davis 的中位數估計量

```
HDmed=function(data){
n=length(data);
b=seq(from=0,to=1,by=1/n); #訂組界 i/n
cb=pbeta(b,(n+1)/2,(n+1)/2); #計算 Beta 分配落在(i-1)/n 和 i/n 之間的機率
res=sum(diff(cb*sort(data))); #資料的加權和
return(res)}
HDmed(x)
```

##計算平均數的信賴區間

```
> x=rnorm(1000)
> y=abs(x) > 1.645
> sum(y)
[1] 103
> sum(y)/length(x)
[1] 0.103
> n=1000
> xbar=mean(x)
> se=sqrt(var(x)/n)
> za=qnorm(0.975)
> ci=c(xbar-se*za, xbar+se*za)
> ci
[1] -0.01611505 0.01611505
```

一、基本敘述統計量

將連續資料算出基本統計量，如樣本平均數 $\bar{x}$ 、標準差 s 等，可以讓分析者對此筆資料有初步的了解，敘述統計量主要有四種型式：

- (1) 集中趨勢的統計量：平均數、中位數、眾數。
- (2) 位置的統計量：百分位數、四分位數。
- (3) 離散的統計量：全距、標準差、四分位差、變異數。
- (4) 形狀的統計量：偏態係數、峰態係數。

範例一：以下是 10 位學生的身高、體重測量值，計算出 BMI，並且想看看平均數、中位數、百分位數、全距、變異數、標準差、峰度與偏度。

height	weight	BMI
173.00	66.00	22.05
164.00	52.00	19.33
180.00	78.00	24.07
152.00	45.00	19.48
170.00	62.00	21.45
168.00	63.00	22.32
162.00	51.00	19.43
175.00	72.00	23.51
170.00	60.00	20.76
158.00	48.00	19.23

R 程式碼：

```
a <- read.table("d:/bmi.txt", header = T)
(Mean<-apply(a, 2, mean))
(Median<-apply(a, 2, median))
p<- c(0.25, 0.75)
(Quantile <- apply(a, 2, function(x)quantile(x, p)))
Range <-apply(a, 2, range)
t(t(Range)%*%c(-1, 1))
(Var <- apply(a, 2, var))
(Sd <- apply(a, 2, sd))
library(e1071)
(Kurtosis <- apply(a, 2, kurtosis))
(Skewness <- apply(a, 2, skewness))
```

報表解說：

```
> a <- read.table("d:/bmi.txt", header = T)
```

```
> (Mean<-apply(a, 2, mean))
```

height	weight	BMI
167.200	59.700	21.163

```
> (Median<-apply(a, 2, median))
```

height	weight	BMI
169.000	61.000	21.105

```
> p<- c(0.25, 0.75)
```

```
> (Quantile <- apply(a, 2, function(x)quantile(x, p)))
```

	height	weight	BMI
25%	162.50	51.25	19.4425
75%	172.25	65.25	22.2525

```
> Range <-apply(a, 2, range)
```

```
> t(t(Range)%*%c(-1, 1))
```

	height	weight	BMI
[1,]	28	33	4.84

```
> (Var <- apply(a, 2, var))
```

height	weight	BMI
69.733333	114.455556	3.251268

```
> (Sd <- apply(a, 2, sd))
```

height	weight	BMI
8.350649	10.698390	1.803127

試開 URL 'http://cran.cs.pu.edu.tw/bin/windows/contrib/2.8/e1071_1.5-19.zip'

The downloaded packages are in

```
C:\Documents and Settings\Administrator\Local  
Settings\Temp\Rtmpgo1OfF\downloaded_packages
```

```
updating HTML package descriptions
```

```
> library(e1071)
```

```
Loading required package: class
```

```
> (Kurtosis <- apply(a, 2, kurtosis))
```

```
height    weight      BMI  
-1.083806 -1.385964 -1.614164
```

```
> (Skewness <- apply(a, 2, skewness))
```

```
height    weight      BMI  
-0.2838034 0.1955393 0.2788264
```

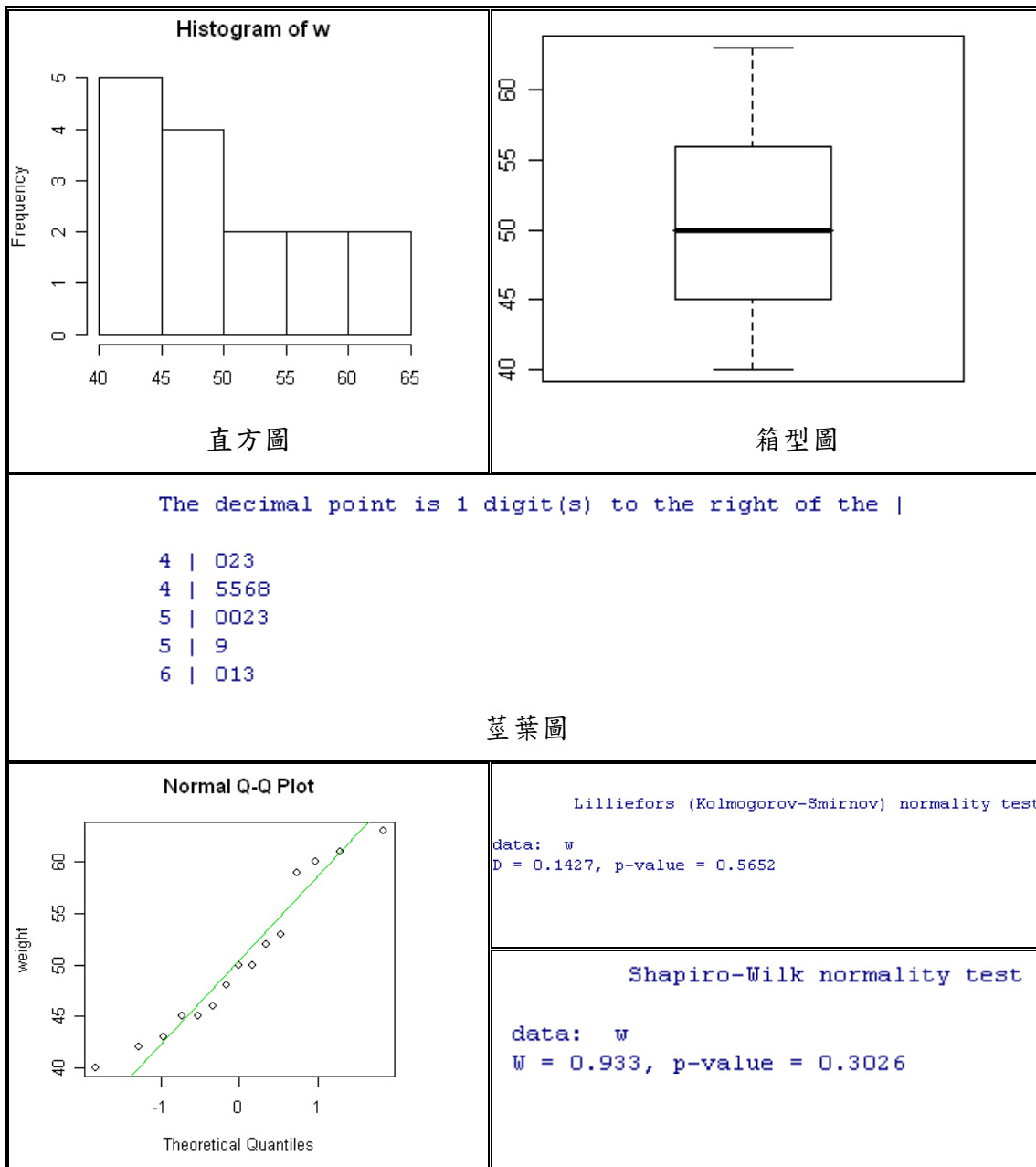
範例二：以下是隨機收集 15 個女生的體重資料，想看看直方圖、莖葉圖、箱型圖，並檢驗此筆資料是否為常態分配。

```
40 42 43 45 45 46 48 50 50 52 53 59 60 61 63
```

R 程式碼：

```
w <- c(40, 42, 43, 45, 45, 46, 48, 50, 50, 52, 53, 59, 60, 61, 63)  
##直方圖  
hist(w)  
##莖葉圖  
stem(w)  
##箱型圖  
boxplot(w)  
##NormalQQ-Plot  
qqnorm(w, ylab = "weight")  
qqline(w, col = 3)  
##檢驗是否常態分配  
library(nortest)  
lillie.test(w)  
shapiro.test(w)
```


報表解說：



根據 Normal Q-Q Plot 圖形顯示，並沒有嚴重違反一條直線，且常態性檢定表，K-S 與 S-W 檢定 P-Value 都大於 0.05，所以假設此筆資料服從常態分配。

二、T 檢定及變異數分析

(1) T 檢定的分類

➤ 單一樣本 t 檢定

$$\begin{cases} H_0: \mu = \mu_0 \\ H_1: \mu \neq \mu_0 \end{cases}, \text{檢定統計量: } t = \frac{\bar{X} - \mu_0}{\frac{S_n}{\sqrt{n}}}, \text{ reject } H_0 \text{ if } |T| > t_{\alpha/2, n-1}$$

➤ 成對 t 檢定

$$* \begin{cases} H_0: \mu_1 = \mu_2 \\ H_1: \mu_1 \neq \mu_2 \end{cases} \Rightarrow \begin{cases} H_0: \mu_d = 0 \\ H_1: \mu_d \neq 0 \end{cases}, \text{檢定統計量: } t = \frac{\bar{D} - 0}{\frac{S_d}{\sqrt{n}}}, \text{ reject } H_0 \text{ if } |T| > t_{\alpha/2, n-1}$$

$$* \begin{cases} H_0: \mu_1 - \mu_2 = d_0 \\ H_1: \mu_1 - \mu_2 \neq d_0 \end{cases}, \text{檢定統計量: } t = \frac{\bar{D} - d_0}{\frac{S_d}{\sqrt{n}}}, \text{ reject } H_0 \text{ if } |T| > t_{\alpha/2, n-1}$$

➤ 獨立 t 檢定

先檢定變異數是否相等，再依情形做檢定：

$$\begin{cases} H_0: \sigma_1^2 = \sigma_2^2 \\ H_1: \sigma_1^2 \neq \sigma_2^2 \end{cases}, \text{檢定統計量: } F = \frac{S_1^2}{S_2^2} \square F_{m-1, n-1}$$

* 變異數相等 ($\sigma_1^2 = \sigma_2^2$)

$$\begin{cases} H_0: \mu_1 - \mu_2 = d_0 \\ H_1: \mu_1 - \mu_2 \neq d_0 \end{cases}, \text{檢定統計量: } t = \frac{\bar{X} - \bar{Y} - d_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \text{ 其中 } S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2},$$

reject H_0 if $|T| > t_{\alpha/2, n_1 + n_2 - 2}$

* 變異數不相等 ($\sigma_1^2 \neq \sigma_2^2$):

$$\begin{cases} H_0: \mu_1 - \mu_2 = d_0 \\ H_1: \mu_1 - \mu_2 \neq d_0 \end{cases}, \text{檢定統計量: } t = \frac{\bar{X} - \bar{Y} - d_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}, \text{ } v(\text{檢定自由度}) = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}},$$

reject H_0 if $|T| > t_{\alpha/2, v}$

● 單一樣本 T 檢定

範例：統計學老師相信統計學期中考的成績平均分數不低於 87 分，現在由考卷中抽取

15 筆資料，檢驗平均分數是否不低於 87 分。

98 97 95 94 90 88 86 84 82 81 79 75 74 70 65

$H_0: u \geq 87$

$H_1: u < 87$

R 程式碼：

```
score <- read.table("d:/test1.txt", header = T)
t.test(score, alternative = "less", mu = 87)
```

報表解說：

```
One Sample t-test

data:  score
t = -1.2059, df = 14, p-value = 0.1239
alternative hypothesis: true mean is less than 87
95 percent confidence interval:
 -Inf 88.44306
sample estimates:
mean of x
 83.86667
```

由上表可知統計學期中成績平均數為 83.87 分，單尾檢定的 P-Value = 0.124，所以統計學期中成績未顯著低於 87 分。

●成對 T 檢定

資料：15 位成年男子參與一個研究計畫，低脂肪食物和有氧運動對降低血液中的膽固醇是否有效果，以下是實驗前和參與計畫三個月後膽固醇的含量：

	1	2	3	4	5	6	7
實驗前	265	240	258	295	251	245	287
實驗後	229	231	230	240	238	241	234

	8	9	10	11	12	13	14	15
實驗前	314	260	279	283	240	238	225	247
實驗後	256	247	239	246	218	219	226	233

R 程式碼：

```
difference <- read.table("d:/test2.txt", header = T)
attach(difference)
t.test(before, after, alternative="greater", mu=0, paired=TRUE)
```

報表解說：

```
Paired t-test

data:  before and after
t = 5.434, df = 14, p-value = 4.403e-05
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 18.02322      Inf
sample estimates:
mean of the differences
      26.66667
```

$H_0: u \geq 0$ (實驗前 - 實驗後 < 0 , 表示無顯著效果)

$H_1: u < 0$ (實驗前 - 實驗後 ≥ 0 , 表示有顯著效果)

由上表可知實驗前-實驗後平均數為 26.67，由 Paired t-test 檢定的表中可看出

P-Value 很小 (< 0.0001)，表示低脂肪食物和有氧運動對降低血液中膽固醇有顯著效果。

●獨立 T 檢定

資料：X 理論和 Y 理論是管理控制的重要理論，甲公司總經理為了解何種理論較有效，

隨機選擇了 16 位員工分成 A、B 兩組，其中 A 組施以 X 理論的環境；

B 組施以 Y 理論的環境，然後在年終給予員工績效評分如下，試檢定兩種管理方式之效果是否有差異。

	1	2	3	4	5	6	7	8
A 組	86	82	84	83	84	83	85	87
B 組	83	81	84	72	79	85	78	86

R 程式碼：

```
effect <- read.table("d:/test3.txt", header = T)
attach(effect)
var.test(A, B, ratio=1, alternative="two.sided")
t.test(A, B, alternative="two.sided", mu=0, var.equal=FALSE)
```

報表解說：

```
F test to compare two variances

data:  A and B
F = 0.1318, num df = 7, denom df = 7, p-value = 0.01583
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.02637821 0.65811304
sample estimates:
ratio of variances
 0.1317568
```

首先，看變異數等值的假設檢定：P-Value = 0.0158，代表變異數不相等。

```
Welch Two Sample t-test

data:  A and B
t = 1.8792, df = 8.813, p-value = 0.09362
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.6750189  7.1750189
sample estimates:
mean of x mean of y
 84.25      81.00
```

$$H_0: u_1 - u_2 = 0$$

$$H_1: u_1 - u_2 \neq 0$$

A 組的平均績效為 84.25，B 組的平均績效為 81，由變異數不等的 Welch t 檢定值：t = 1.88，P-Value = 0.0936 > 0.05，所以可知 X 理論與 Y 理論的效果無顯著差異。

(2)變異數分析

變異數分析(ANOVA)的目的在於找出自變項與依變項之間的關係，或是自變項對依變項產生的實驗效果。簡言之，即為檢驗兩組以上母體的平均數是否有顯著差異時，就需使用變異數分析。

* 基本模型

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij} \quad i = 1, \dots, k \quad j = 1, \dots, n_i$$

* 模型假設

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$$

* 統計假設

$$\begin{cases} H_0: \mu_1 = \mu_2 = \dots = \mu_k \\ H_1: \mu_i \text{ 不完全相等} \end{cases}$$

$$* \text{ 總平方和: } SSTo = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_{..})^2$$

$$* \text{ 處理(組間)平方和: } SStr(SSb) = \sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{X}_{i.} - \bar{X}_{..})^2$$

$$* \text{ 誤差(組內)平方和: } SSE(SSw) = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_{i.})^2$$

$$\Rightarrow SSTo = SStr + SSE, \quad MStr = \frac{SStr}{k-1}, \quad MSE = \frac{SSE}{n-k}$$

ANOVA Table

變異來源	平方和	自由度	均方和	F 值
處理	SStr	k-1	$MStr = \frac{SStr}{k-1}$	$F = \frac{MStr}{MSE}$
誤差	SSE	n-k	$MSE = \frac{SSE}{n-k}$	
總計	SSTo	n-1		

●One-Way ANOVA

資料：針對某企業內的五個單位，各抽樣 5 位員工的年終考績分數，如下表，試計算此五個單位員工的年終考績是某相同。

生產(1)	5	4	8	6	3
行銷(2)	9	7	8	6	9
研發(3)	3	5	2	3	7
人事(4)	2	3	4	1	4
總務(5)	7	6	9	4	7

R 程式碼：

```
worker<- read.table("d:/test4.txt", header = T)
attach(worker)
fit <- lm(score~factor(group))
summary.aov(fit)
TukeyHSD(aov(score~factor(group)), conf.level=0.95)
```

報表解說：

```
          Df Sum Sq Mean Sq F value    Pr(>F)
factor(group)  4   79.44    19.86   6.8958 0.001170 **
Residuals    20   57.60     2.88
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

根據 ANOVA 表 P-Value = 0.0012 可知五個單位員工的年終考績顯著不相同，以下進行多重比較。

```
Tukey multiple comparisons of means
95% family-wise confidence level

Fit: aov(formula = score ~ factor(group))

$`factor(group)`
      diff      lwr      upr      p adj
2-1  2.6 -0.6117534  5.8117534 0.1500459
3-1 -1.2 -4.4117534  2.0117534 0.7952722
4-1 -2.4 -5.6117534  0.8117534 0.2072026
5-1  1.4 -1.8117534  4.6117534 0.6916608
3-2 -3.8 -7.0117534 -0.5882466 0.0156173
4-2 -5.0 -8.2117534 -1.7882466 0.0012742
5-2 -1.2 -4.4117534  2.0117534 0.7952722
4-3 -1.2 -4.4117534  2.0117534 0.7952722
5-3  2.6 -0.6117534  5.8117534 0.1500459
5-4  3.8  0.5882466  7.0117534 0.0156173
```

由 Tukey 事後檢定可知，信賴區間沒有包含 0，表示兩平均數有顯著差異，所以行銷(2)與研發(3)、人事(4)考績分數有顯著不同，人事(4)與總務(5)考績分數有顯著不同

三、卡方檢定

類別資料分析是針對反應變數為類別變數資料(解釋變數可能是連續或類別變數)的一種分析方法。類別資料分析的標準方法包括兩類別變數的相關性評估(卡方獨立性檢定)，以及解釋變數對類別反應變數關聯的影響(羅吉斯迴歸分析)。

●卡方適合度檢定

檢定所蒐集到的資料其觀察的次數與理論推導出來的期望次數是否接近。統計量：

$$\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$$

範例：一家民調公司以電話訪查 450 位晚間無線電視新聞之觀眾，詢問其最愛喜愛的

新聞頻道，結果為台視 158 位、中視 102 位、華視 91 位及民視 99 位，請問四家無線電視台市場佔有率是否有差異。

台視 A	中視 B	華視 C	民視 D
158	102	91	99

R 程式碼：

```
freq <- matrix(c(158,102,91,99),nrow=1)
dimnames(freq) <- list("Freq", c("1","2","3","4"))
chisq.test(freq)
```

報表解說：

$$\begin{cases} H_0 : P_1 = P_2 = P_3 = P_4 = \frac{1}{4} \\ H_1 : \text{at least one } P_i \neq \frac{1}{4} \end{cases}$$

```
Chi-squared test for given probabilities

data:  freq
X-squared = 25.1111, df = 3, p-value = 1.464e-05
```

由卡方檢定結果 P-Value < 0.0001，代表四家無線電視台市場佔有率有顯著差異。

●獨立性檢定

檢定同個母體中兩個變數是否獨立。

$$\begin{cases} H_0 : A \text{ 與 } B \text{ 獨立} \\ H_1 : A \text{ 與 } B \text{ 不獨立} \end{cases}$$

$$\chi^2 = \sum_{i=1}^a \sum_{j=1}^b \frac{(o_{ij} - e_{ij})^2}{e_{ij}} \sim \chi_{(a-1)(b-1)}^2$$

$$\text{reject } H_0 \text{ if } \chi^2 > \chi_{(a-1)(b-1), \alpha}^2$$

範例：為判斷工作績效(A 因素)與收入高低(B 因素)間的關係，某公司觀察結果如下，試問兩者是否具有關聯性。

	0~2 萬	2~4 萬	4~6 萬	6~8 萬
績效低	12	13	11	4
績效高	9	14	21	16

R 程式碼：


```
work <- matrix(c(12, 13, 11, 4, 9, 14, 21, 16), byrow=TRUE, nrow=2)
dimnames(work) <- list(efficacy=c("low", "high"), revenue=c("0~2", "2~4", "4~6",
                                                             "6~8"))
work
chisq.test(work)
```

報表解說：

```

              revenue
efficacy 0~2 2~4 4~6 6~8
low      12  13  11   4
high      9  14  21  16

```

根據二維的列聯表，收入在 0~2 萬的且工作績效低的有 12 人

收入在 0~2 萬的且工作績效高的有 9 人

收入在 2~4 萬的且工作績效低的有 13 人

收入在 2~4 萬的且工作績效高的有 14 人

收入在 4~6 萬的且工作績效低的有 11 人

收入在 4~6 萬的且工作績效高的有 21 人

收入在 6~8 萬的且工作績效低的有 4 人

收入在 6~8 萬的且工作績效高的有 16 人

Pearson's Chi-squared test

```
data: work
X-squared = 7.0736, df = 3, p-value = 0.06959
```

由皮爾森卡方檢定 P-Value = 0.07，顯示收入高低與工作績效無顯著相關。

四、迴歸分析

迴歸分析的主要目的是作預測，目標是發展一種能以一個或多個預測變數的數值來做為預測反應變數的方法。

迴歸分析的前提假設：

- (1) 反應變數 Y 服從常態分配，且變異數相等。
- (2) 觀察值之間是獨立的，解釋變數 X 之間也是獨立的。
- (3) 解釋變數 X 與反應變數 Y 為線性關係。

(4) 殘差之間獨立且服從常態分配。

* 簡單線性迴歸模式

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad i=1, 2, \dots, n \quad \text{where } \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

$$\Rightarrow y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

* 線性複迴歸模式

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_k x_{ki} + \varepsilon_i \quad i=1, 2, \dots, n \quad \text{where } \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

● 簡單線性迴歸分析

範例：下表是 10 位學生的期中和期末成績，想用迴歸分析以期中成績去預測期末成績。

期末成績	79	71	80	67	65	68	78	68	58	57
期中成績	71	61	69	72	60	55	71	61	57	56

R 程式碼：

```
score <- read.table("d:/test5.txt", header = T)
attach(score)
fit <- lm(final~midterm)
summary(fit)
anova(fit)
plot(fit, 1, add.smooth=F)
plot(fit, 2)
```

報表解說：

```
Call:
lm(formula = final ~ midterm)

Residuals:
    Min     1Q   Median     3Q    Max
-9.900 -4.374  1.479  3.721  6.342

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  12.3446   18.0211   0.685   0.5127
midterm       0.8966    0.2832   3.165   0.0133 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

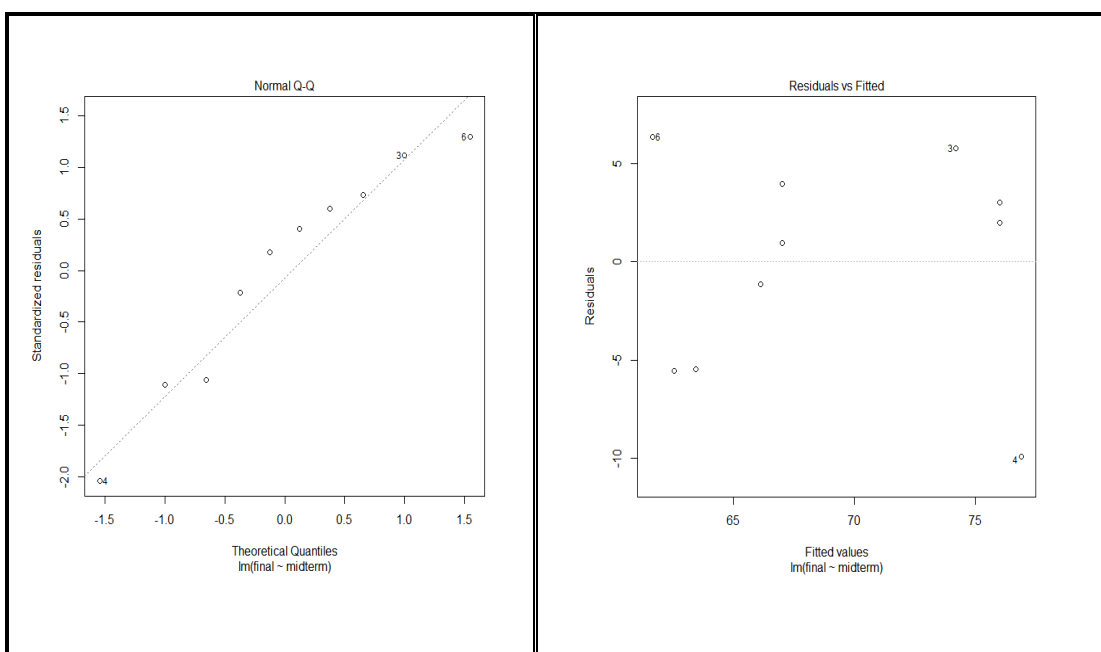
Residual standard error: 5.736 on 8 degrees of freedom
Multiple R-squared:  0.5561,    Adjusted R-squared:  0.5006
F-statistic: 10.02 on 1 and 8 DF, p-value: 0.01329
```

此迴歸模式可寫成：期末成績 = 12.34 + 0.897 * 期中成績 + 殘差值。模型的判定係數，通常看 R Square，此模型解釋變數(期中成績)對反應變數(期末成績)有 55.6% 的解釋能力。

Analysis of Variance Table					
Response: final					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
midterm	1	329.68	329.68	10.02	0.01329 *
Residuals	8	263.22	32.90		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

變異數分析表的 P-Value = 0.013，表示此模型的變數具有解釋能力(即此解釋變數與反應變數具相關性)，是一個有效的迴歸公式。



根據 Normal Q-Q Plot 圖形顯示，並沒有嚴重違反一條直線，殘差圖形分散程度尚可，符合迴歸的前提假設。

●線性複迴歸分析

範例：個人所得(每月)受到年齡、教育程度、工作經驗的影響，隨機收集 10 戶資料如下，試作 y 對 x_1, x_2, x_3 的複迴歸分析。

序號	個人所得(萬元) Y	年齡(歲) X_1	教育程度(年) X_2	工作經驗(年) X_3
1	2	19	12	1
2	2.5	23	14	2

3	3	25	14	2
4	3.3	26	15	3
5	3.8	27	15	3
6	4.5	27	15	4
7	5	29	16	5
8	7	31	17	6
9	9	33	17	6
10	11	35	17	8

R 程式碼：

```
revenue <- read.table("d:/test6.txt", header = T)
attach(revenue)
fit <- lm(money~age+level+work)
summary(fit)
anova(fit)
plot(fit, 1 , add.smooth=F)
plot(fit, 2)
fit2 <- lm(money~work)
summary(fit2)
anova(fit2)
plot(fit2, 1 , add.smooth=F)
plot(fit2, 2)
```

報表解說：

```

Call:
lm(formula = money ~ age + level + work)

Residuals:
    Min       1Q   Median       3Q      Max
-0.8980 -0.2652 -0.1548  0.3935  1.1120

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.1956     5.0837   1.219   0.2687
age           0.5943     0.2823   2.105   0.0799 .
level        -1.4178     0.6019  -2.356   0.0566 .
work          1.0309     0.4087   2.522   0.0451 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7193 on 6 degrees of freedom
Multiple R-Squared: 0.9611,    Adjusted R-squared: 0.9416
F-statistic: 49.35 on 3 and 6 DF,  p-value: 0.0001273

```

先將三個解釋變數都放入迴歸模式中，所建立出的模型為

$\hat{y} = 6.196 + 0.594x_1 - 1.418x_2 + 1.031x_3$ ，模型的判定係數，通常看 Adjust R Square，此模型解釋變數對反應變數有 94.16%的解釋能力。P-Value = 0.0001，表示此模型的變數具有解釋能力(即此解釋變數與反應變數具相關性)，是一個有效的迴歸公式，但從係數的顯著性可看出有些解釋變數不顯著，所以剔除不顯著的變數。

```

Call:
lm(formula = money ~ work)

Residuals:
    Min       1Q   Median       3Q      Max
-1.40318 -0.58670 -0.02023  0.65705  1.30364

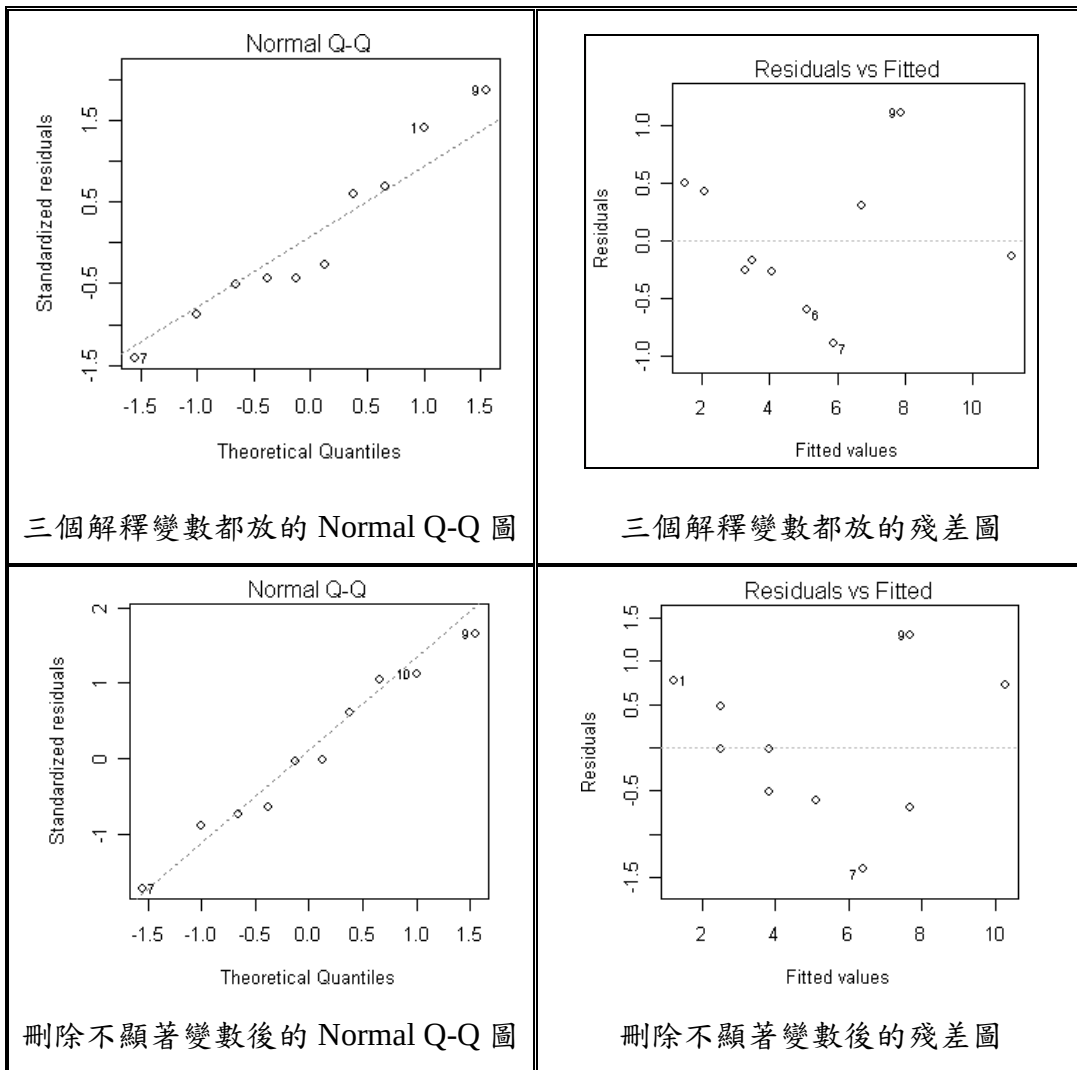
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.06273     0.59589  -0.105   0.919
work         1.29318     0.13193   9.802 9.85e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8751 on 8 degrees of freedom
Multiple R-Squared: 0.9231,    Adjusted R-squared: 0.9135
F-statistic: 96.08 on 1 and 8 DF,  p-value: 9.854e-06

```

挑選出的變數為工作經驗，排除的變數為年齡與教育程度，所以可以建立出另一模型為

$\hat{y} = -0.063 + 1.293x_3$ ，此模型的解釋能力為 91.35%。P-value < 0.0001，表示此模型的變數具有解釋能力(即此解釋變數與反應變數具相關性)，是一個有效的迴歸公式。



由 Normal Q-Q 圖和殘差圖可知，刪除不顯著變數後的 Normal Q-Q 圖更接近一條直線，且殘差圖也比較均勻分散。

五、無母數統計方法

當我們要討論的是母體分配未知且樣本數不是很大時，就需要利用無母數的方法來處理，此方法不限定欲推論母體之分配，通常利用百分位(如：中位數)，或是排序的方式來分析資料。其依樣本的多寡可分為：一組樣本的統計推論、二組獨立樣本的統計推論、二組相關樣本的統計推論、k 組獨立樣本的統計推論、k 組相關樣本的統計推論。但無母數方法的缺點為：檢定力較弱、處理方式不一和檢定查表的表格很多。

●One-Sample Kolmogorov-Smirnov Test

範例：以下是隨機收集 15 個女生的體重資料，想檢驗此筆資料是否為常態分配。

41 42 43 45 45 46 48 50 50 52 53 59 60 61 63

R 程式碼：

```
library(nortest)
w <- c(41, 42, 43, 45, 45, 46, 48, 50, 50, 52, 53, 59, 60, 61, 63)
lillie.test(w)
```

報表解說：

```
Lilliefors (Kolmogorov-Smirnov) normality test

data:  w
D = 0.144, p-value = 0.5507
```

根據常態性檢定表，K-S 檢定 P-Value = 0.5507，所以假設此筆資料服從常態分配。

●Mann-Whitney Test

範例：以下為 A、B 兩生產線之產品的長度資料，分別自該兩生產線抽取 6 個隨機樣本，其長度(公分)如下表，試以 Mann-Whitney 檢定兩生產線產品之平均長度是否有差異？

生產線 A	138	138	140	144	132	136
生產線 B	144	140	138	142	134	142

R 程式碼：

```
x <- c(138, 138, 140, 144, 132, 136)
y <- c(144, 140, 138, 142, 134, 142)
wilcox.test(x, y, alternative = "two.sided")
```

報表解說：

```
Wilcoxon rank sum test with continuity correction

data:  x and y
W = 12, p-value = 0.3726
alternative hypothesis: true location shift is not equal to 0
```

由上表可知，威爾森正準檢定 P-value = 0.3726，代表兩生產線產品平均長度無顯著差異。

●Kruskal-Wallis Test

範例：某校使用三種不同教學方法所得學生之成就測驗成績如下，試利用 K-W 和 Medain

Test 檢定不同教學法對於學生成就測驗成績是否有影響？

啟發式(x)	86	84	89	91	84	82
CAI 式(y)	91	89	92	85	88	91
傳統式(z)	85	82	75	72	74	82

R 程式碼：

```
x <- c(86,84,89,91,84,82)
y <- c(91,89,92,85,88,91)
z <- c(85,82,75,72,74,82)
kruskal.test(list(x, y, z),alternative = "two.sided")
```

報表解說：

```
Kruskal-Wallis rank sum test

data: list(x, y, z)
Kruskal-Wallis chi-squared = 10.5792, df = 2, p-value = 0.005044
```

由 Kruskal-Wallis Test 所得 P-Value = 0.005，代表三種不同教學方法對於學生成就測驗之成績有顯著影響。

●Random Sampling

範例：使用 bmi1.txt 資料檔，以取後不放回隨機抽取 5 筆資料和以取後放回隨機抽取 15 筆資料，分別計算變數 height(身高)和 weight(體重)之線性相關係數。

R 程式碼：

```
data <- scan("d:/bmi1.txt")
a <- matrix(data, 10, 3, byrow=T)
height <- a[,1]
weight <- a[,2]
b <- a[,1:2]
# sampling without replacement
c <- b[sample(1:10, 5),]
c1 <- cor(height, weight)
c2 <- cor(c[,1],c[,2])
a
c
c1
```



```

c2
# sampling with replacement
d <- b[sample(1:10, size=15,replace=TRUE),]
d
d1 <- cor(d[,1],d[,2])
d1

```

報表解説：

```

> a
      [,1] [,2] [,3]
[1,]  173   66 22.05
[2,]  164   52 19.33
[3,]  180   78 24.07
[4,]  152   45 19.48
[5,]  170   62 21.45
[6,]  168   63 22.32
[7,]  162   51 19.43
[8,]  175   72 23.51
[9,]  170   60 20.76
[10,] 158   48 19.23

> c
      [,1] [,2]
[1,]  152   45
[2,]  158   48
[3,]  162   51
[4,]  164   52
[5,]  168   63

> c1

```

```
[1] 0.9671083
```

```
> c2
```

```
[1] 0.904506
```

```
>d
```

```
      [,1] [,2]
```

```
[1,] 173   66
```

```
[2,] 168   63
```

```
[3,] 158   48
```

```
[4,] 162   51
```

```
[5,] 173   66
```

```
[6,] 168   63
```

```
[7,] 152   45
```

```
[8,] 170   62
```

```
[9,] 170   60
```

```
[10,] 168   63
```

```
[11,] 173   66
```

```
[12,] 180   78
```

```
[13,] 152   45
```

```
[14,] 170   62
```

```
[15,] 170   62
```

```
>d1
```

```
[1] 0.9671083
```

HW: 隨機生成 100 筆常態分配 $N(5,4)$ 資料，並計算其基本敘述統計量。

再將資料編號成 1, ..., 100，自此 100 筆資料以取後不放回和以取後放回各隨機抽

取 30 筆資料，分別計算其基本敘述統計量。

●Data Generation

範例：(1)隨機生成 30 筆標準常態分配資料，並計算其基本敘述統計量。

(2)隨機生成 30 筆均勻分配資料，並計算其基本敘述統計量。

(3)隨機生成 30 筆指數分配資料，並計算其基本敘述統計量。

R 程式碼：

```
data1 <- matrix(rnorm(30), 30, 1)
a <- summary(data1)
a
library(e1071)
(Kurtosis <- apply(data1, 2, kurtosis))
(Skewness <- apply(data1, 2, skewness))

title <- paste("Histogram")
par(mfrow=c(1,2))
hist(data1)
qqnorm(data1)
qqline(data1)
```

報表解說：

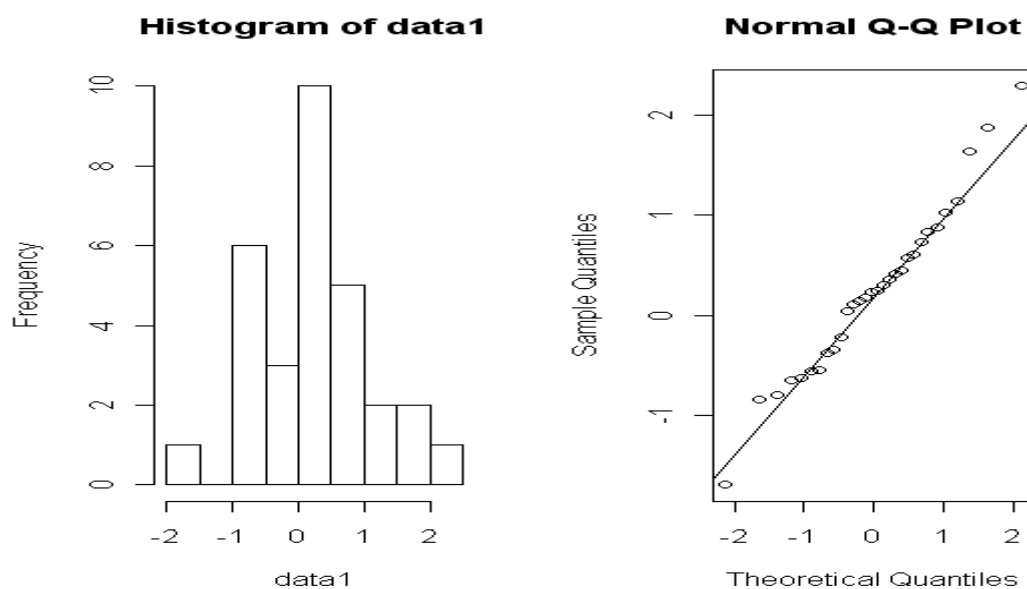
```
V1
Min.   :-1.6991
1st Qu.: -0.3680
Median :  0.2402
Mean    :  0.2469
3rd Qu.:  0.6983
Max.    :  2.2930

> library(e1071)
Loading required package: class
> (Kurtosis <- apply(data1, 2, kurtosis))
```

```
[1] -0.005137176
```

```
> (Skewness <- apply(data1, 2, skewness))
```

```
[1] 0.2453681
```



R 程式碼：

```
data2 <- matrix(runif(30), 30, 1)
b <- summary(data2)
b
(Kurtosis <- apply(data2, 2, kurtosis))
(Skewness <- apply(data2, 2, skewness))
hist(data2)
qqnorm(data2)
qqline(data2)
```

報表解說：

V1

Min. :0.07344

1st Qu.:0.28920

Median :0.48587

Mean :0.49713

3rd Qu.:0.69119

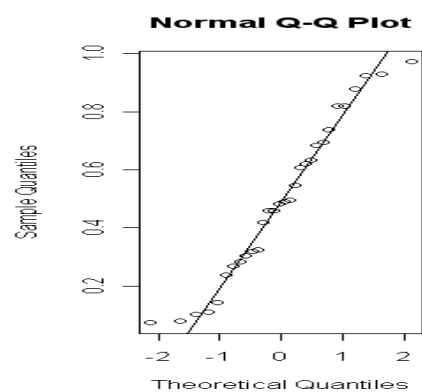
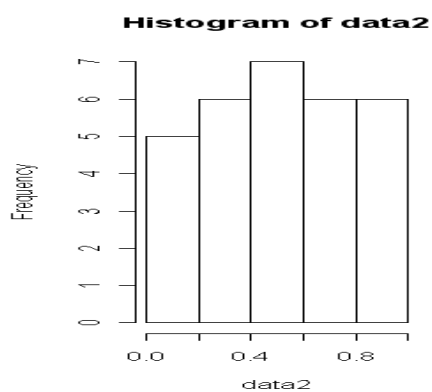
Max. :0.97173

```
> (Kurtosis <- apply(data2, 2, kurtosis))
```

```
[1] -1.202633
```

```
> (Skewness <- apply(data2, 2, skewness))
```

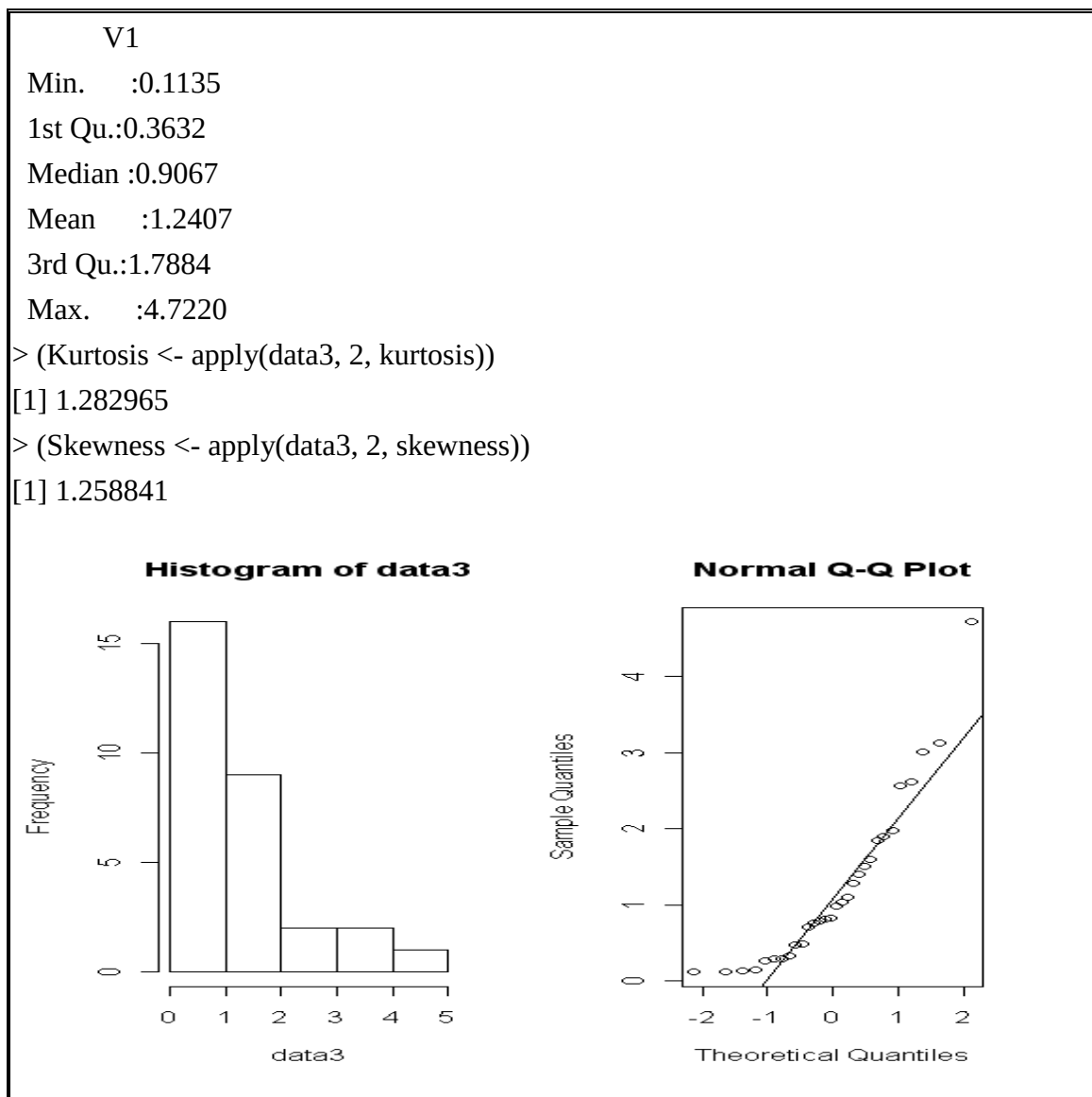
```
[1] 0.080131
```



R 程式碼：

```
data3 <- matrix(rexp(30), 30, 1)
c <- summary(data3)
(Kurtosis <- apply(data3, 2, kurtosis))
(Skewness <- apply(data3, 2, skewness))
par(mfrow=c(1,2))
hist(data3)
qqnorm(data3)
qqline(data3)
```

報表解說：



●Data Analysis

範例 1：利用套裝軟體 R 中的資料檔 trees，進行敘述統計分析，並檢定資料是否呈常態分配。

R 程式碼及報表：

```

> data(trees)
> str(trees)
'data.frame':   31 obs. of  3 variables:
 $ Girth : num   8.3 8.6 8.8 10.5 10.7 10.8 11 11 11.1 11.2 ...
 $ Height: num   70 65 63 72 81 83 66 75 80 75 ...
 $ Volume: num  10.3 10.3 10.2 16.4 18.8 19.7 15.6 18.2 22.6 19.9 ...
> options(digits=4)
> var(trees)
      Girth Height Volume
Girth   9.848  10.38  49.89
Height 10.383  40.60  62.66
Volume 49.888  62.66 270.20
> sd(trees)
      Girth Height Volume
3.138   6.372 16.438
> options(digits=7)
> var(trees)
      Girth   Height   Volume
Girth  9.847914 10.38333  49.88812
Height 10.38333 40.60000  62.66000
Volume 49.888118 62.66000 270.20280
> summary(trees)
      Girth           Height           Volume
Min.   : 8.30   Min.   :63   Min.   :10.20
1st Qu.:11.05   1st Qu.:72   1st Qu.:19.40
Median :12.90   Median :76   Median :24.20
Mean   :13.25   Mean   :76   Mean   :30.17
3rd Qu.:15.25   3rd Qu.:80   3rd Qu.:37.30
Max.   :20.60   Max.   :87   Max.   :77.00
> sapply(trees,IQR)
      Girth Height Volume
4.2      8.0  17.9
> round(sapply(trees, function(x) sd(x)/mean(x)),4)
      Girth Height Volume
0.2369 0.0838 0.5448
> fivenum(trees)
[1]  8.3 14.0 24.2 72.0 87.0
> stem(trees[,1])

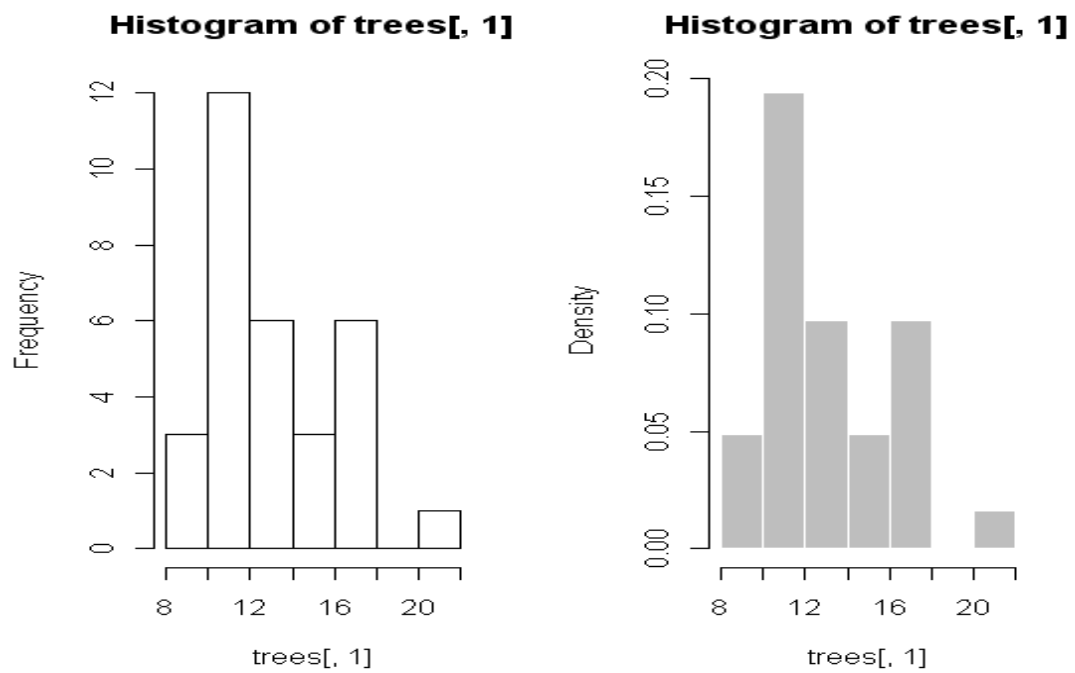
```

```

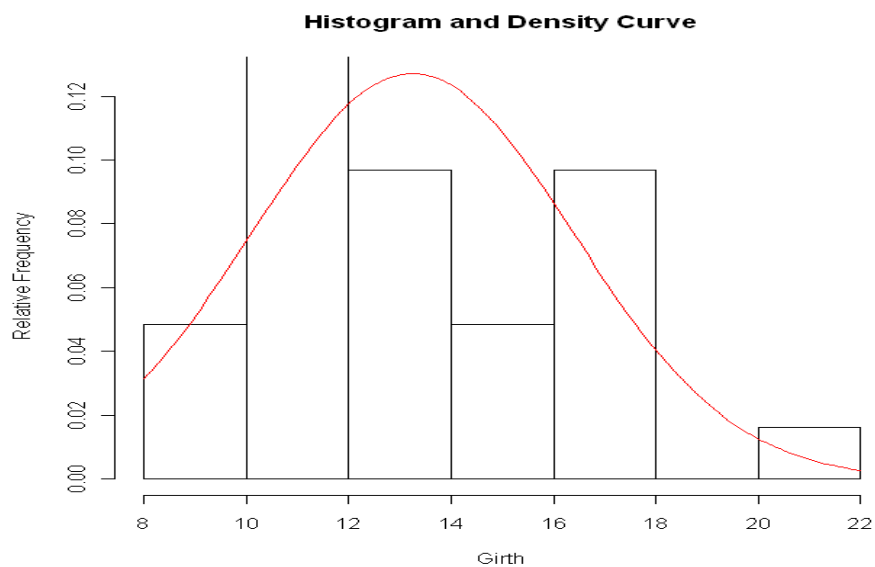
The decimal point is at the |
  8 | 368
10 | 57800123447
12 | 099378
14 | 025
16 | 03359
18 | 00
20 | 6
> stem(trees[,1],scale=2)
The decimal point is at the |
  8 | 368
  9 |
10 | 578
11 | 00123447
12 | 099
13 | 378
14 | 025
15 |
16 | 03
17 | 359
18 | 00
19 |
20 | 6
> shapiro.test(log(trees$Volume))
      Shapiro-Wilk normality test
data:  log(trees$Volume)
W = 0.9643, p-value = 0.3766

> par(mfrow=c(1,2))
> hist(trees[,1])
> hist(trees[,1],freq=FALSE,col="gray", border="white")

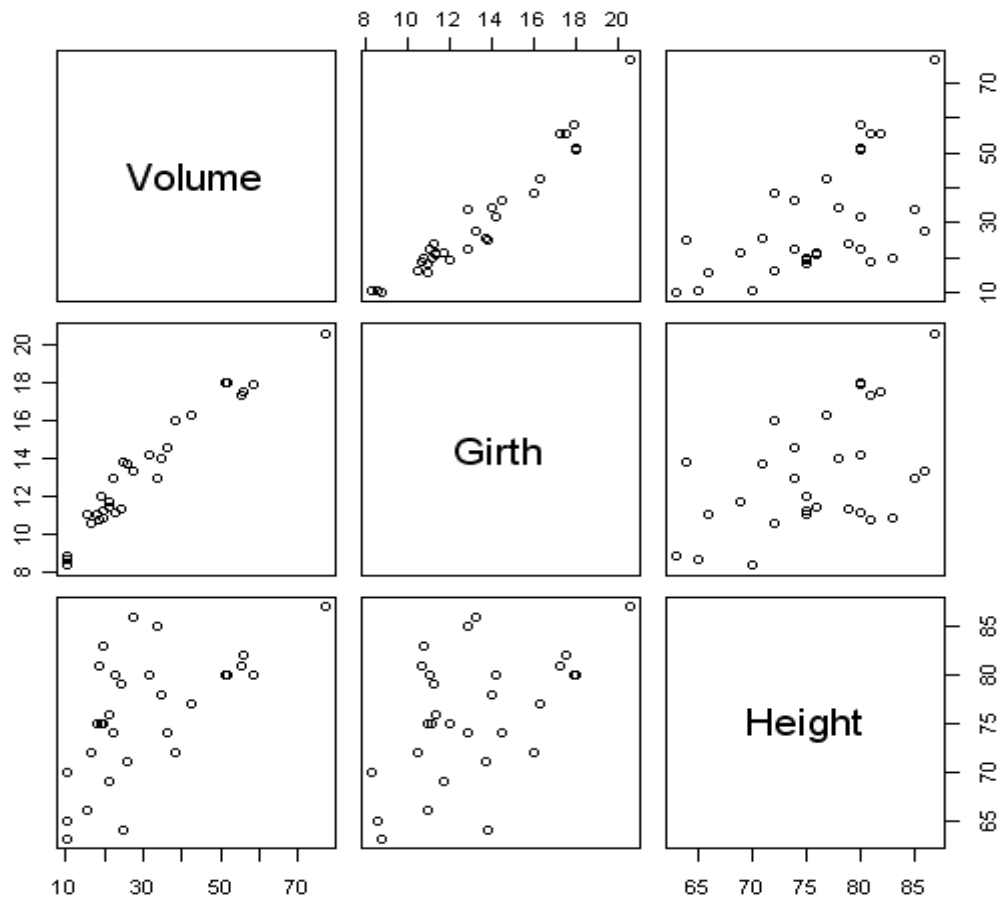
```

```
mu <- mean(trees[, 1])
sigma <- sd(trees[, 1])
hist(trees[, 1], freq=FALSE, ylim=c(0, dnorm(mu, mu, sigma)),
xlab="Girth", ylab="Relative Frequency", main="Histogram and Density Curve")
curve(dnorm(x, mu, sigma), add=TRUE, col=2)
```

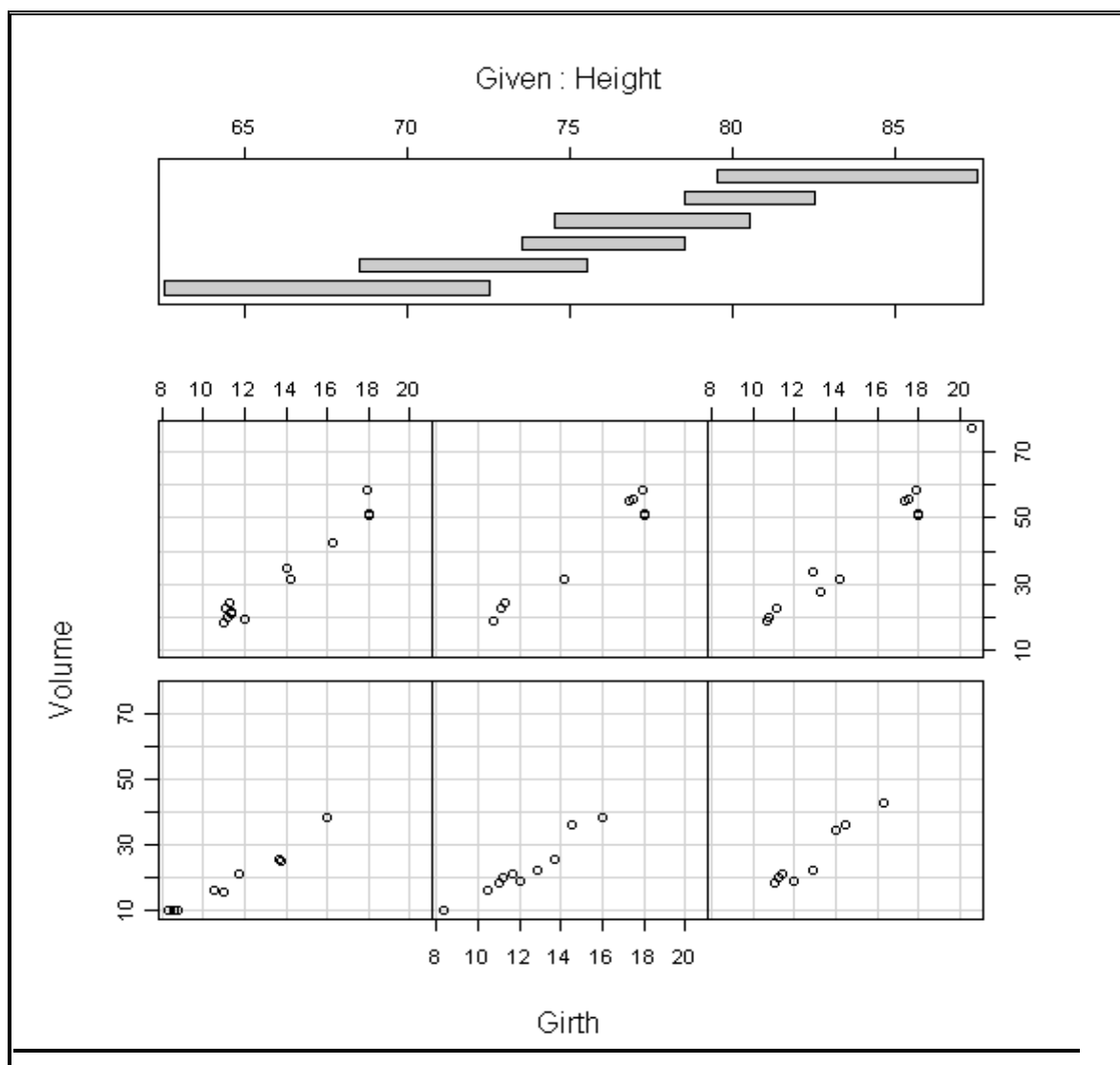


```
pairs(Volume~Girth+Height, data=trees)
```



```
> attach(trees)
```

```
> coplot(Volume~Girth|Height, data=trees)
```



範例 2：利用套裝軟體 R 中的資料檔 UCBAmissions，作交叉分析表。

R 程式碼及報表：

```
data(UCBAmissions)
```

```
ftable(UCBAmissions)
```

		Dept					
Admit	Gender	A	B	C	D	E	F
Admitted	Male	512	353	120	138	53	22
	Female	89	17	202	131	94	24
Rejected	Male	313	207	205	279	138	351
	Female	19	8	391	244	299	311

範例 3：利用套裝軟體 R 中的資料檔 warpbreaks，進行敘述統計分析，並畫長條圖 (bar chart)和盒鬚圖(box plot)。

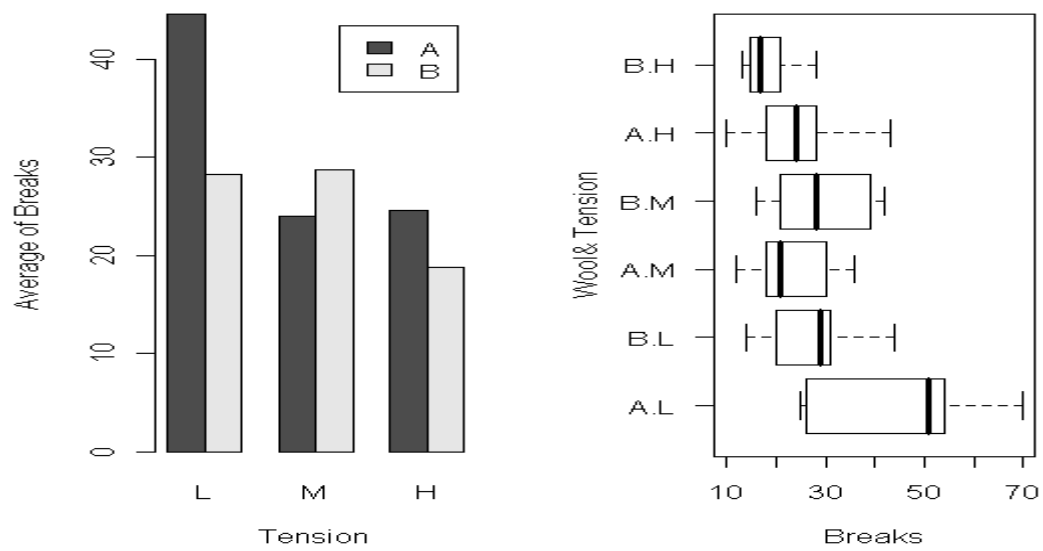
R 程式碼及報表：

```
> attach(warpbreaks)
> str(warpbreaks)
'data.frame':  54 obs. of  3 variables:
 $ breaks : num  26 30 54 25 70 52 51 26 67 18 ...
 $ wool   : Factor w/ 2 levels "A","B": 1 1 1 1 1 1 1 1 1 1 ...
 $ tension: Factor w/ 3 levels "L","M","H": 1 1 1 1 1 1 1 1 1 2 ...

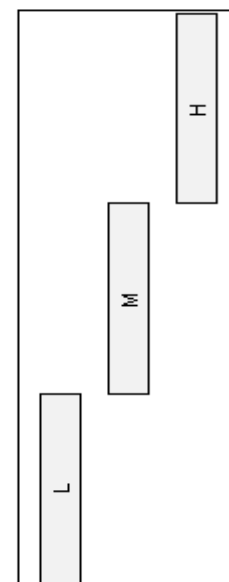
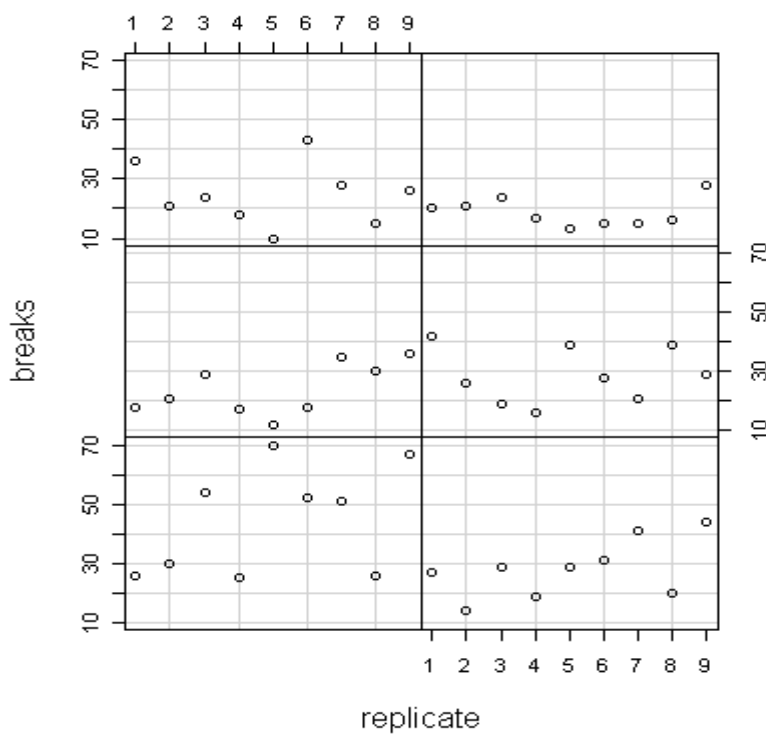
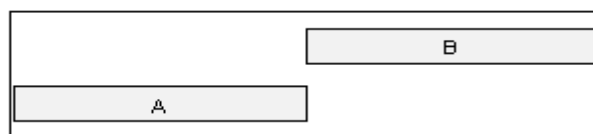
> summary(warpbreaks)
      breaks      wool      tension
Min.   :10.00   A:27    L:18
1st Qu.:18.25   B:27    M:18
Median :26.00                H:18
Mean    :28.15
3rd Qu.:34.00
Max.    :70.00

par(mfrow=c(1,2))
with(warpbreaks,{
  barplot(tapply(breaks, list(wool, tension), mean),
    beside=TRUE, legend=TRUE, ylab="Average of Breaks",
    xlab="Tension",
    main=paste("Breaks in Yarn During Weaving",
      "Wool Type:A and B", sep="\n"))
  boxplot(breaks~wool:tension, horizontal=TRUE, las=1,
    xlab="Breaks", ylab="Wool& Tension",
    main=paste("Breaks in Yarn During Weaving",
      "Box Plot", sep="\n"))
})
> replicate <- factor(rep(1:9,6))
> N <- length(breaks)
> k <- length(levels(wool:tension))
> replicate <- gl(N/k,1,N)
> str(replicate)
Factor w/ 9 levels "1","2","3","4",...: 1 2 3 4 5 6 7 8 9 1 ...
> coplot(breaks~replicate|wool+tension)
```

Breaks in Yarn During Weaving **Breaks in Yarn During Weaving**
Wool Type:A and B **Box Plot**



Given : wool



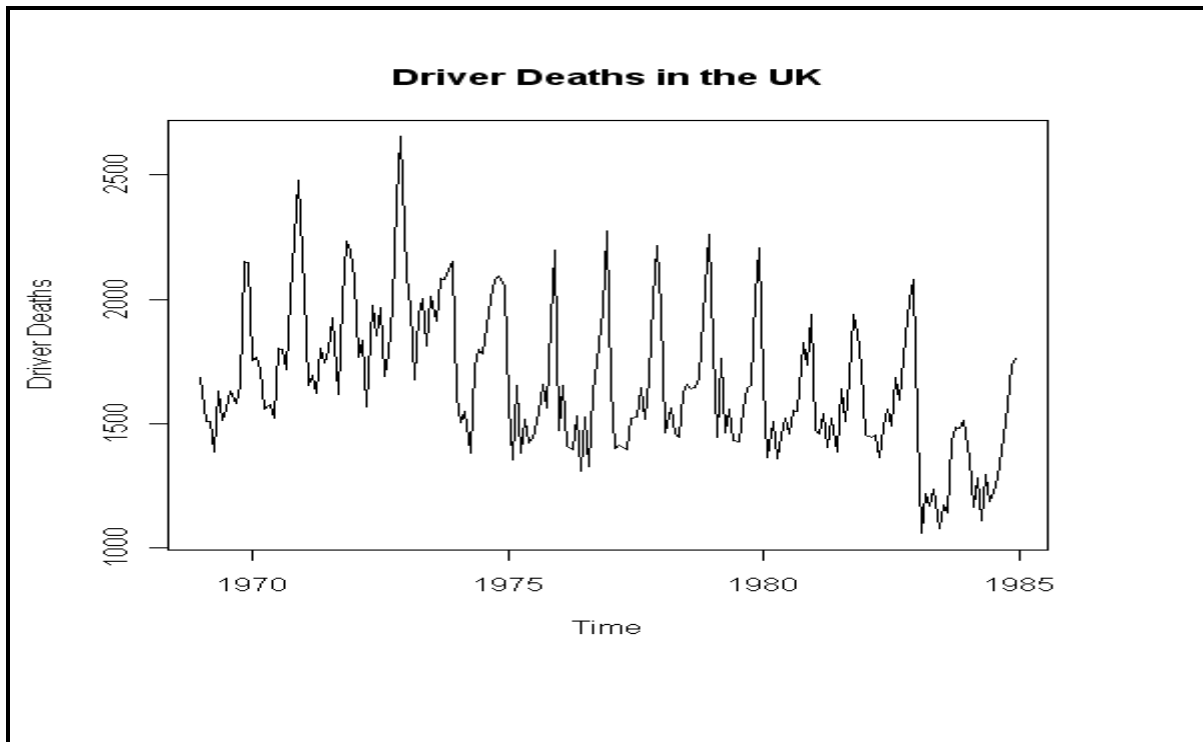
Given : tension

範例 4:利用套裝軟體 R 中的資料檔 UKDriverDeaths，畫時間數列圖(time series plot)。

R 程式碼：

```
data(UKDriverDeaths)
str(UKDriverDeaths)
Time-Series [1:192] from 1969 to 1985: 1687 1508 1507 1385 1632 ...
plot(UKDriverDeaths, ylab="Driver Deaths", main="Driver Deaths in the UK")
```

報表：



範例 5:利用套裝軟體 R 中的資料檔 Nile 和 sunspots，畫時間數列圖(time series plot)。

R 程式碼：

```
layout(matrix(c(1,1,2,2,1,1,3,3), 2, 4, byrow=TRUE))
data(Nile)
plot(Nile, ylab="Annual Flow", main="The River Nile at Ashwan")
data(sunspots)
plot(sunspot.year, main="Sunspots: 1970-1988")
plot(sunspot.month, main="Sunspots: 1949-1997")
layout(1)

help(layout)
##layout divides the device up into as many rows and columns as there are in matrix mat,
with the column-widths and the row-heights specified in the respective arguments.
```

```

def.par <- par(no.readonly = TRUE) # save default, for resetting...

## divide the device into two rows and two columns
## allocate figure 1 all of row 1
## allocate figure 2 the intersection of column 2 and row 2
layout(matrix(c(1,1,0,2), 2, 2, byrow = TRUE))
## show the regions that have been allocated to each plot
layout.show(2)

## divide device into two rows and two columns
## allocate figure 1 and figure 2 as above
## respect relations between widths and heights
nf <- layout(matrix(c(1,1,0,2), 2, 2, byrow=TRUE), respect=TRUE)
layout.show(nf)

## create single figure which is 5cm square
nf <- layout(matrix(1), widths=lcm(5), heights=lcm(5))
layout.show(nf)

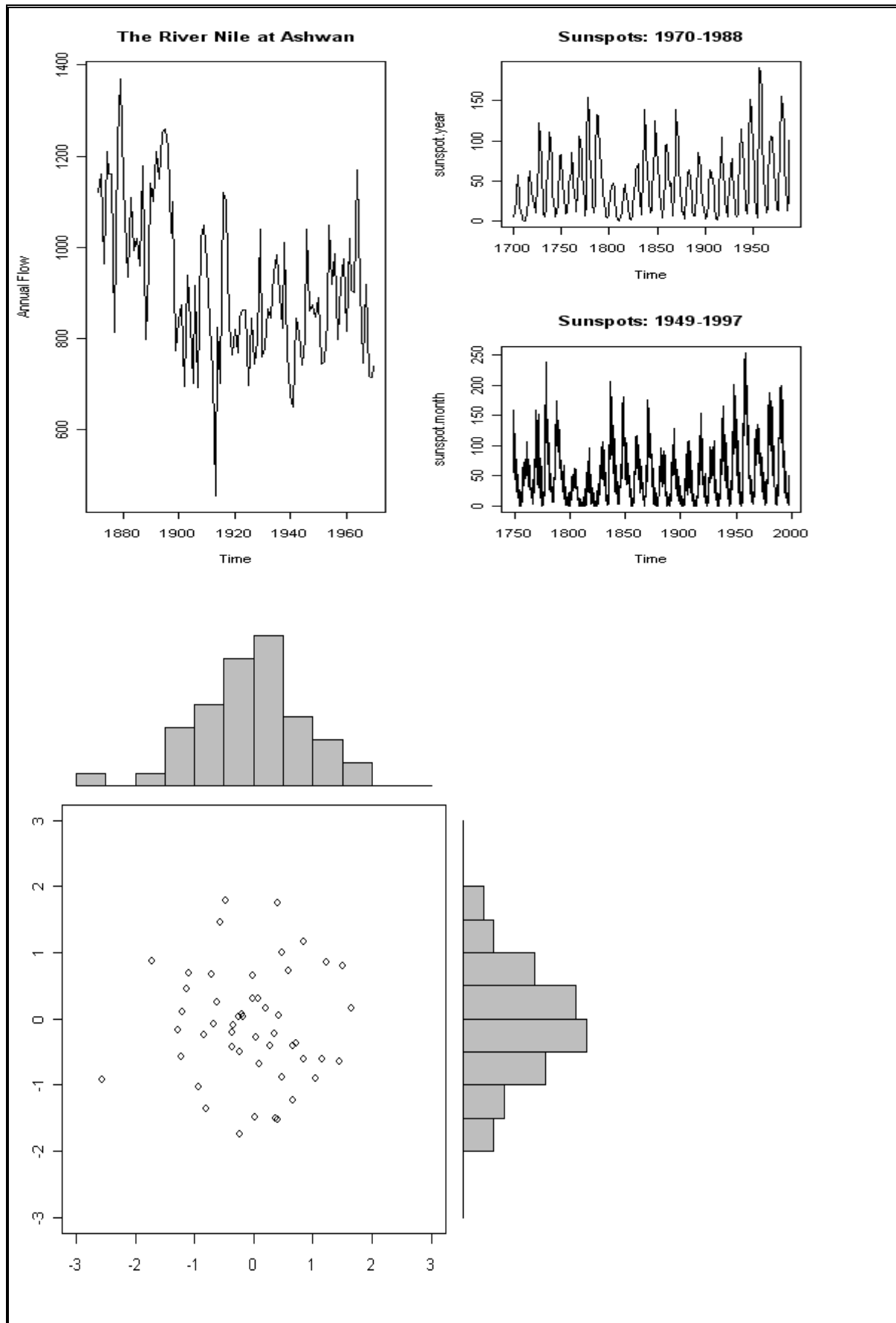
##-- Create a scatterplot with marginal histograms -----

x <- pmin(3, pmax(-3, stats::rnorm(50)))
y <- pmin(3, pmax(-3, stats::rnorm(50)))
xhist <- hist(x, breaks=seq(-3,3,0.5), plot=FALSE)
yhist <- hist(y, breaks=seq(-3,3,0.5), plot=FALSE)
top <- max(c(xhist$counts, yhist$counts))
xrange <- c(-3,3)
yrange <- c(-3,3)
nf <- layout(matrix(c(2,0,1,3),2,2,byrow=TRUE), c(3,1), c(1,3), TRUE)
layout.show(nf)

par(mar=c(3,3,1,1))
plot(x, y, xlim=xrange, ylim=yrange, xlab="", ylab="")
par(mar=c(0,3,1,1))
barplot(xhist$counts, axes=FALSE, ylim=c(0, top), space=0)
par(mar=c(3,0,1,1))
barplot(yhist$counts, axes=FALSE, xlim=c(0, top), space=0, horiz=TRUE)
par(def.par)#- reset to default

```

報表：



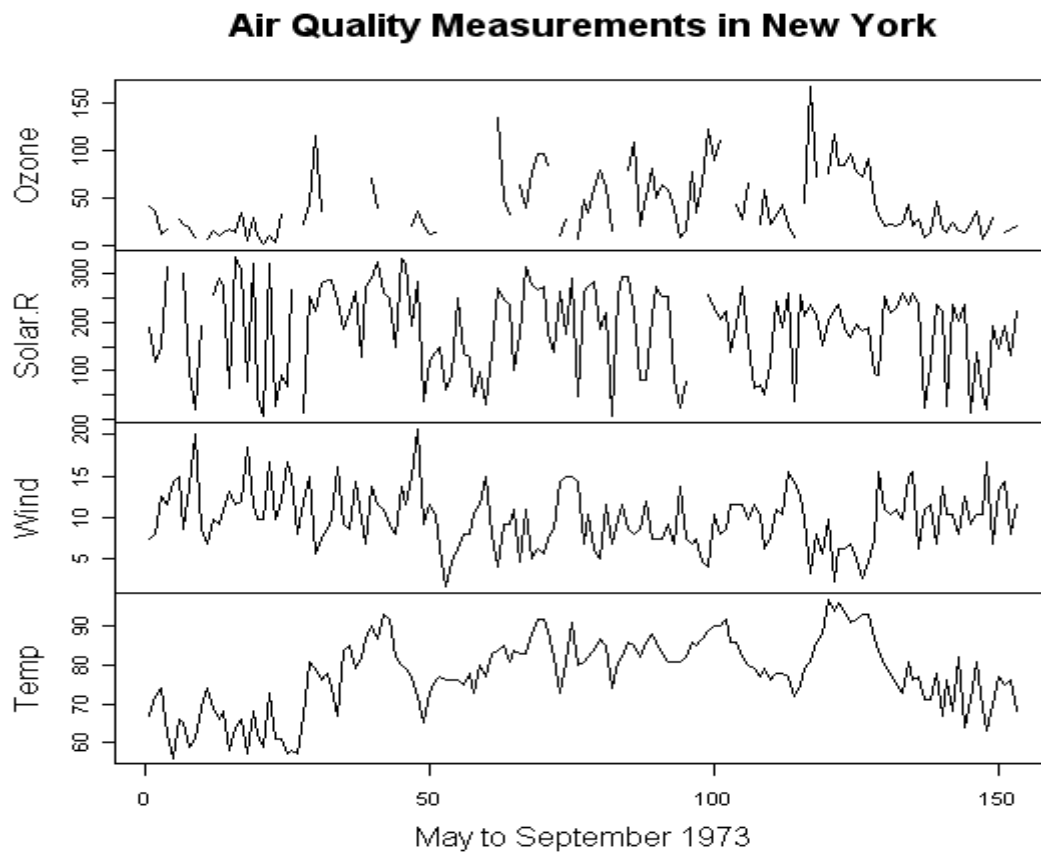
範例 6：利用套裝軟體 R 中的資料檔 airquality，畫時間數列圖(time series plot)。

R 程式碼：

```
data(airquality)
str(airquality)
plot.ts(airquality[, 1:4], xlab="May to September 1973",
main="Air Quality Measurements in New York")
```

報表：

```
'data.frame':  153 obs. of  6 variables:
 $ Ozone   : int  41 36 12 18 NA 28 23 19 8 NA ...
 $ Solar.R: int  190 118 149 313 NA NA 299 99 19 194 ...
 $ Wind    : num   7.4 8 12.6 11.5 14.3 14.9 8.6 13.8 20.1 8.6 ...
 $ Temp    : int   67 72 74 62 56 66 65 59 61 69 ...
 $ Month    : int   5 5 5 5 5 5 5 5 5 5 ...
 $ Day     : int   1 2 3 4 5 6 7 8 9 10 ...
```

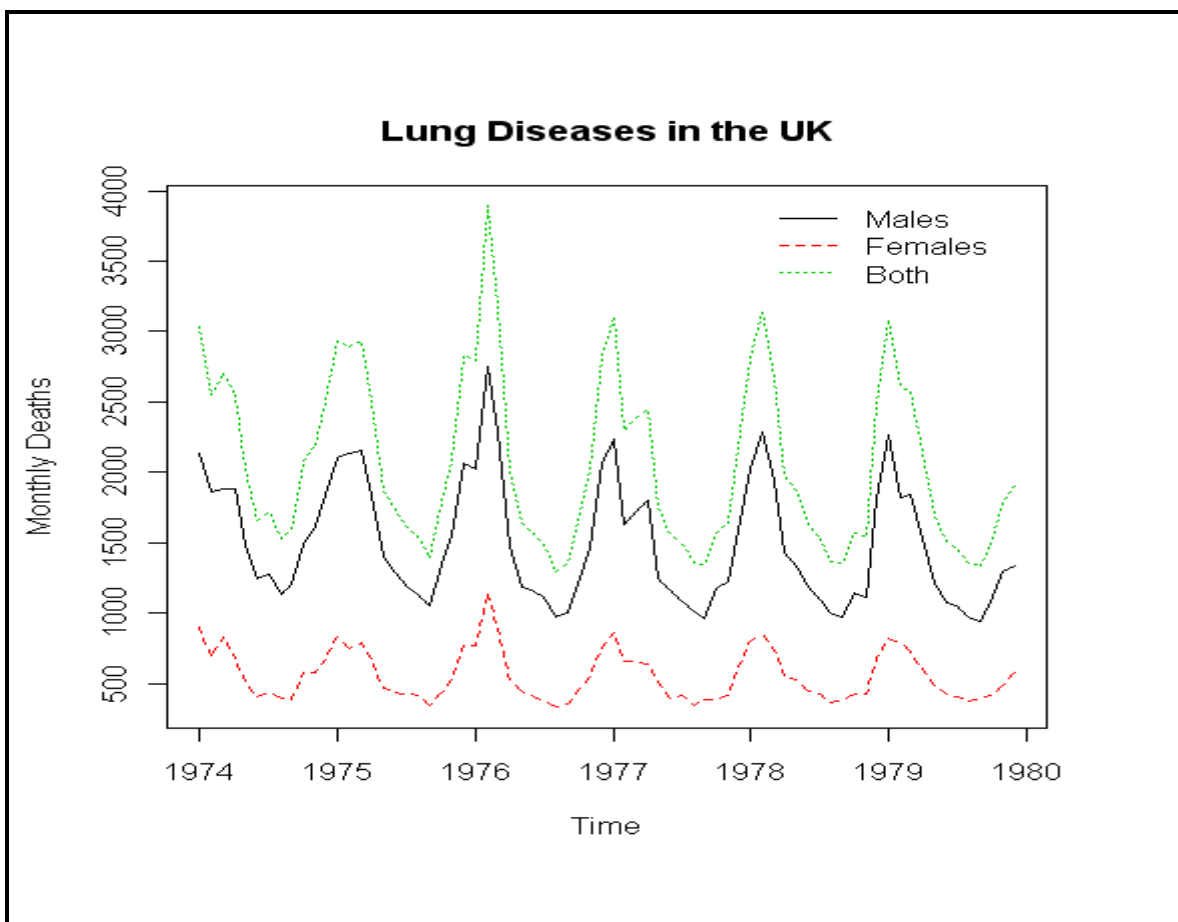


範例 7：利用套裝軟體 R 中的資料檔 UKLungDeaths，畫時間數列圖(time series plot)。

R 程式碼：

```
data(UKLungDeaths)
exists(c("ldeaths","mdeaths","fdeaths"))
plot(cbind(mdeaths, fdeaths, ldeaths), plot.type="single",
lty=4:6, col=3:5, ylab="Monthly Deaths", main="Lung Diseases in the UK")
legend(1974, 4000, c("Males","Females","Both"),
lty=4:6, col=3:5, bty="o")
```

報表：



範例：找出套裝軟體 R 中所有的資料檔

```
> data()
```

Data sets in package ‘datasets’:

AirPassengers	Monthly Airline Passenger Numbers 1949-1960
BJsales	Sales Data with Leading Indicator
BJsales.lead (BJsales)	Sales Data with Leading Indicator
BOD	Biochemical Oxygen Demand
CO2	Carbon Dioxide uptake in grass plants
ChickWeight	Weight versus age of chicks on different diets
DNase	Elisa assay of DNase
EuStockMarkets	Daily Closing Prices of Major European Stock Indices, 1991-1998
Formaldehyde	Determination of Formaldehyde
HairEyeColor	Hair and Eye Color of Statistics Students
Harman23.cor	Harman Example 2.3
Harman74.cor	Harman Example 7.4
Indometh	Pharmacokinetics of Indomethicin
InsectSprays	Effectiveness of Insect Sprays
JohnsonJohnson	Quarterly Earnings per Johnson & Johnson Share
LakeHuron	Level of Lake Huron 1875-1972
LifeCycleSavings	Intercountry Life-Cycle Savings Data
Loblolly	Growth of Loblolly pine trees
Nile	Flow of the River Nile
Orange	Growth of Orange Trees
OrchardSprays	Potency of Orchard Sprays
PlantGrowth	Results from an Experiment on Plant Growth
Puromycin	Reaction velocity of an enzymatic reaction
Seatbelts	Road Casualties in Great Britain 1969-84
Theoph	Pharmacokinetics of theophylline
Titanic	Survival of passengers on the Titanic
ToothGrowth	The Effect of Vitamin C on Tooth Growth in Guinea Pigs
UCBAdmissions	Student Admissions at UC Berkeley
UKDriverDeaths	Road Casualties in Great Britain 1969-84
UKgas	UK Quarterly Gas Consumption

USAccDeaths	Accidental Deaths in the US 1973-1978
USArrests	Violent Crime Rates by US State
USJudgeRatings	Lawyers' Ratings of State Judges in the US Superior Court
USPersonalExpenditure	Personal Expenditure Data
VADeaths	Death Rates in Virginia (1940)
WWWusage	Internet Usage per Minute
WorldPhones	The World's Telephones
ability.cov	Ability and Intelligence Tests
airmiles	Passenger Miles on Commercial US Airlines, 1937-1960
airquality	New York Air Quality Measurements
anscombe	Anscombe's Quartet of "Identical" Simple Linear Regressions
attenu	The Joyner-Boore Attenuation Data
attitude	The Chatterjee-Price Attitude Data
austres	Quarterly Time Series of the Number of Australian Residents
beaver1 (beavers)	Body Temperature Series of Two Beavers
beaver2 (beavers)	Body Temperature Series of Two Beavers
cars	Speed and Stopping Distances of Cars
chickwts	Chicken Weights by Feed Type
co2	Mauna Loa Atmospheric CO2 Concentration
crimtab	Student's 3000 Criminals Data
discoveries	Yearly Numbers of Important Discoveries
esoph	Smoking, Alcohol and (O)esophageal Cancer
euro	Conversion Rates of Euro Currencies
euro.cross (euro)	Conversion Rates of Euro Currencies
eurodist	Distances Between European Cities
faithful	Old Faithful Geyser Data
fdeaths (UKLungDeaths)	Monthly Deaths from Lung Diseases in the UK
freeny	Freeny's Revenue Data
freeny.x (freeny)	Freeny's Revenue Data
freeny.y (freeny)	Freeny's Revenue Data
infert	Infertility after Spontaneous and Induced Abortion
iris	Edgar Anderson's Iris Data

iris3	Edgar Anderson's Iris Data
islands	Areas of the World's Major Landmasses
ldeaths (UKLungDeaths)	Monthly Deaths from Lung Diseases in the UK
lh	Luteinizing Hormone in Blood Samples
longley	Longley's Economic Regression Data
lynx	Annual Canadian Lynx trappings 1821-1934
mdeaths (UKLungDeaths)	Monthly Deaths from Lung Diseases in the UK
morley	Michaelson-Morley Speed of Light Data
mtcars	Motor Trend Car Road Tests
nhtemp	Average Yearly Temperatures in New Haven
nottem	Average Monthly Temperatures at Nottingham, 1920-1939
occupationalStatus	Occupational Status of Fathers and their Sons
precip	Annual Precipitation in US Cities
presidents	Quarterly Approval Ratings of US Presidents
pressure	Vapor Pressure of Mercury as a Function of Temperature
quakes	Locations of Earthquakes off Fiji
randu	Random Numbers from Congruential Generator RANDU
rivers	Lengths of Major North American Rivers
rock	Measurements on Petroleum Rock Samples
sleep	Student's Sleep Data
stack.loss (stackloss)	Brownlee's Stack Loss Plant Data
stack.x (stackloss)	Brownlee's Stack Loss Plant Data
stackloss	Brownlee's Stack Loss Plant Data
state.abb (state)	US State Facts and Figures
state.area (state)	US State Facts and Figures
state.center (state)	US State Facts and Figures
state.division (state)	US State Facts and Figures
state.name (state)	US State Facts and Figures
state.region (state)	US State Facts and Figures
state.x77 (state)	US State Facts and Figures
sunspot.month	Monthly Sunspot Data, 1749-1997
sunspot.year	Yearly Sunspot Data, 1700-1988

sunspots	Monthly Sunspot Numbers, 1749-1983
swiss	Swiss Fertility and Socioeconomic Indicators (1888) Data
treering	Yearly Treering Data, -6000-1979
trees	Girth, Height and Volume for Black Cherry Trees
uspop	Populations Recorded by the US Census
volcano	Topographic Information on Auckland's Maunga Whau Volcano
warpbreaks	The Number of Breaks in Yarn during Weaving
women	Average Heights and Weights for American Women

Use ‘data(package = .packages(all.available = TRUE))’
to list the data sets in all *available* packages.

範例：從 Excel file ‘mr.csv’ 讀入整理過的資料檔並重整。

```
zz <- read.csv("d:/mri.csv", strip.white = TRUE)
zzz <- cbind(zz[gl(nrow(zz), 1, 4*nrow(zz)), 1:2], stack(zz[, 3:6]))
zz
  X   Age   V1   V2   V3   V4
1  P 23646 45190 50333 55166 56271
2  CC 26174 35535 38227 37911 41184
3  CC 27723 25691 25712 26144 26398
4  CC 27193 30949 29693 29754 30772
5  CC 24370 50542 51966 54341 54273
6  CC 28359 58591 58803 59435 61292
7  CC 25136 45801 45389 47197 47126
> zzz
  X   Age values ind
1  P 23646 45190 V1
2  CC 26174 35535 V1
3  CC 27723 25691 V1
4  CC 27193 30949 V1
5  CC 24370 50542 V1
6  CC 28359 58591 V1
7  CC 25136 45801 V1
1.1 P 23646 50333 V2
```

```

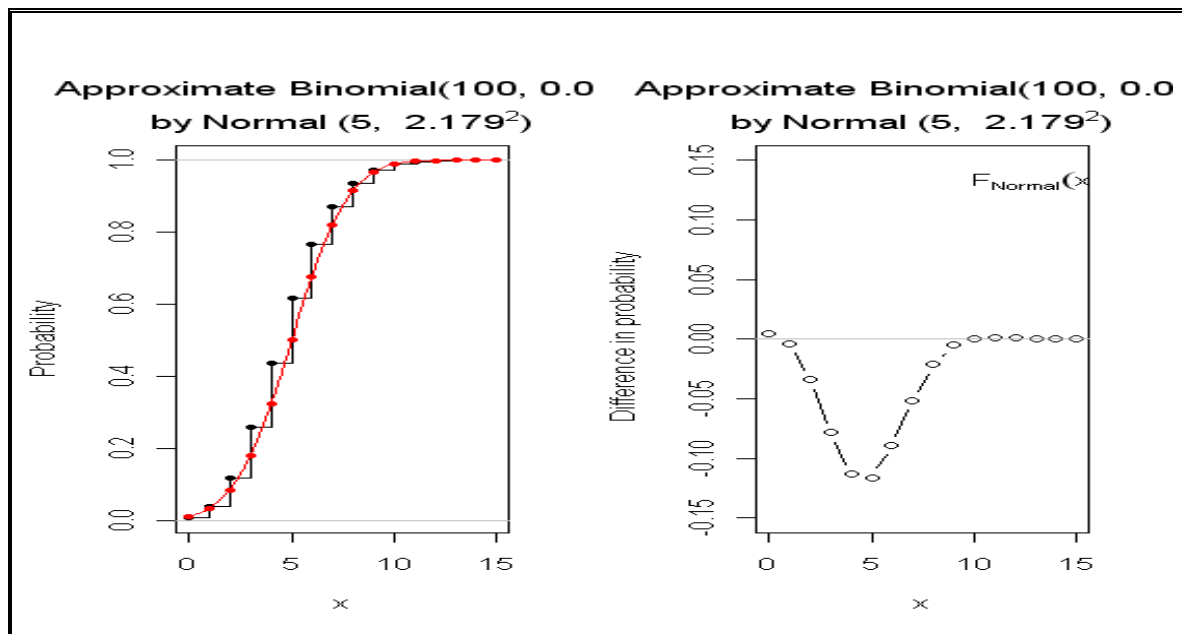
2.1 CC 26174 38227 V2
3.1 CC 27723 25712 V2
4.1 CC 27193 29693 V2
5.1 CC 24370 51966 V2
6.1 CC 28359 58803 V2
7.1 CC 25136 45389 V2
1.2 P 23646 55166 V3
2.2 CC 26174 37911 V3
3.2 CC 27723 26144 V3
4.2 CC 27193 29754 V3
5.2 CC 24370 54341 V3
6.2 CC 28359 59435 V3
7.2 CC 25136 47197 V3
1.3 P 23646 56271 V4
2.3 CC 26174 41184 V4
3.3 CC 27723 26398 V4
4.3 CC 27193 30772 V4
5.3 CC 24370 54273 V4
6.3 CC 28359 61292 V4
7.3 CC 25136 47126 V4
> reshape(zz, idvar="id", timevar="var",
varying=list(c("V1","V2","V3","V4")),direction="long")
      X   Age var    V1 id
1.1 P 23646 1 45190 1
2.1 CC 26174 1 35535 2
3.1 CC 27723 1 25691 3
4.1 CC 27193 1 30949 4
5.1 CC 24370 1 50542 5
6.1 CC 28359 1 58591 6
7.1 CC 25136 1 45801 7
1.2 P 23646 2 50333 1
2.2 CC 26174 2 38227 2
3.2 CC 27723 2 25712 3
4.2 CC 27193 2 29693 4
5.2 CC 24370 2 51966 5
6.2 CC 28359 2 58803 6
7.2 CC 25136 2 45389 7
1.3 P 23646 3 55166 1
2.3 CC 26174 3 37911 2

```

3.3	CC 27723	3 26144	3
4.3	CC 27193	3 29754	4
5.3	CC 24370	3 54341	5
6.3	CC 28359	3 59435	6
7.3	CC 25136	3 47197	7
1.4	P 23646	4 56271	1
2.4	CC 26174	4 41184	2
3.4	CC 27723	4 26398	3
4.4	CC 27193	4 30772	4
5.4	CC 24370	4 54273	5
6.4	CC 28359	4 61292	6
7.4	CC 25136	4 47126	7

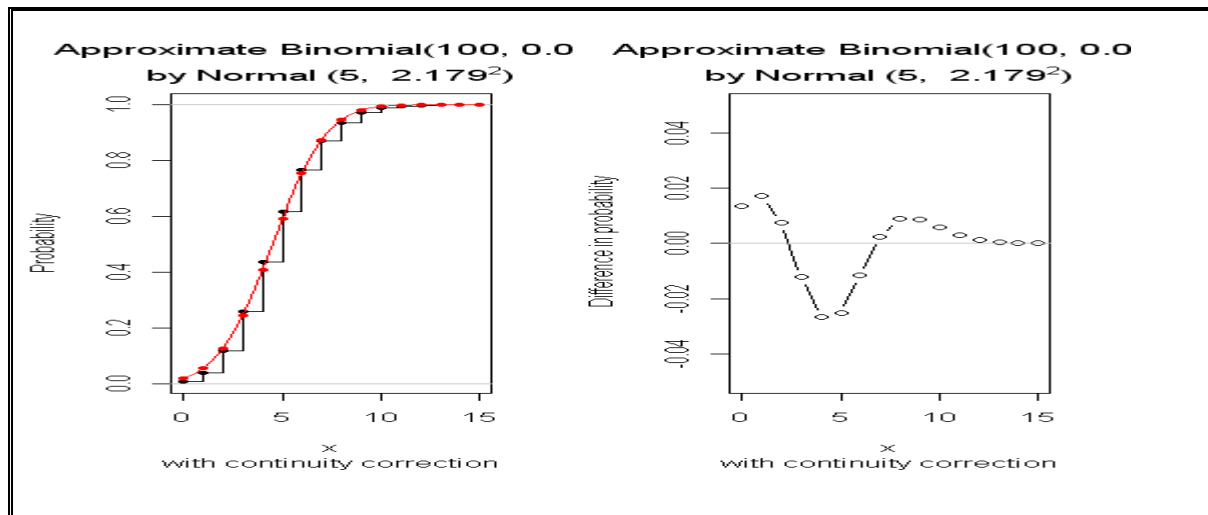
範例：利用常態分配累積分配函數(cdf)近似二項式分配累積分配函數。

```
n <- 100
p <- 0.05
mu <- n*p
sigma <- sqrt(n*p*(1-p))
x <- 0:qbinom(0.9999, n, p)
## general graphic title
main <- substitute(atop("Approximate Binomial(" ~ nn ~ ", " ~ pp ~ ")",
"by Normal (" ~ mm ~ ", " ~ ss ^ 2 ~ ")"),
list(nn=n, pp=p, mm=mu, ss=signif(sigma,4)))
## binomial probability
par(mfrow=c(1,2))
plot(x, pbinom(x, n, p), type="s", ylab="Probability", main=main)
abline(h=0:1, col="gray")
points(x, pbinom(x, n, p), pch=16, cex=0.75)
## add normal density curve
curve(pnorm(x, mu, sigma), add=TRUE, col=2)
points(x, pnorm(x, mu, sigma), col=2, pch=16, cex=0.75)
## difference between normal density and binomial probability
plot(x, pnorm(x, mu, sigma)-pbinom(x, n, p), type="b", ylim=c(-1, 1)*0.15,
ylab=" Difference in probability", main=main)
abline(h=0, col="gray")
legend(2, 0.15, expression(F[Normal](x)-F[Binomial](x)), bty="n")
```

範例：利用常態分配累積分配函數(cdf)近似二項式分配累積分配函數(連續修正)。

```
sub <- "with continuity correction"
## binomial probability
par(mfrow=c(1,2))
plot(x, pbinom(x, n, p), type="s", ylab="Probability", main=main, sub=sub)
abline(h=0:1, col="gray")
points(x, pbinom(x, n, p), pch=16, cex=0.75)
## add normal density curve with continuity correction
mu <- n*p
sigma <- sqrt(n*p*(1-p))
curve(pnorm(x+0.5, mu, sigma), add=TRUE, col=2)
points(x, pnorm(x+0.5, mu, sigma), col=2, pch=16, cex=0.75)
## difference between normal density and binomial probability
plot(x, pnorm(x+0.5, mu, sigma)-pbinom(x, n, p), type="b", ylim=c(-1, 1)*0.05,
ylab=" Difference in probability", main=main, sub=sub)
abline(h=0, col="gray")
legend(7, 0.15, expression(F[Normal](x)-F[Binomial](x)), bty="n")
```

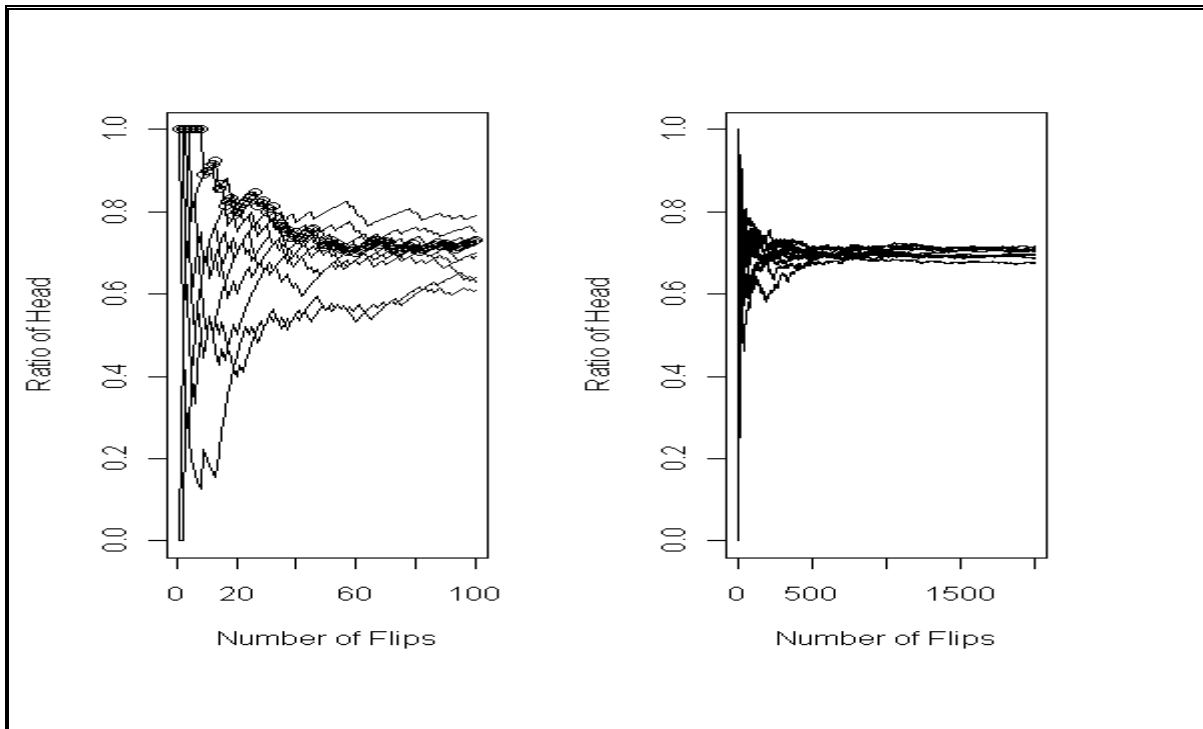


範例：利用二項式分配的樣本平均數驗證大數法則。

```
par(mfrow=c(1,2))
n <- 100
p <- 0.7
nn <- 1:n
x <- cumsum(rbinom(n, 1, p))/nn
plot(x, ylim=c(0,1), type="o",
xlab="Number of Flips", ylab="Ratio of Head")
for (i in 1:10) lines(cumsum(rbinom(n, 1, p))/nn)

n <- 2000
p <- 0.7
nn <- 1:n
plot( cumsum(rbinom(n, 1, p))/nn, ylim=c(0,1), type="l",
xlab="Number of Flips", ylab="Ratio of Head")
for (i in 1:10) lines(cumsum(rbinom(n, 1, p))/nn)

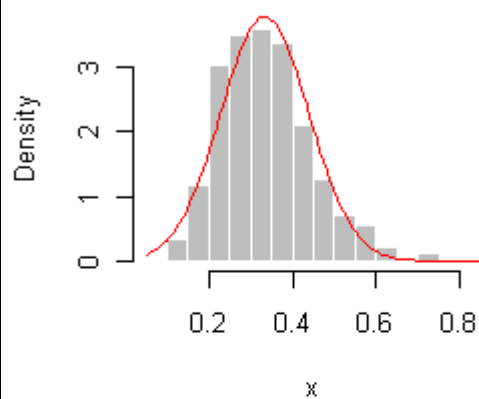
matplot(replicate(11, cumsum(rbinom(2000, 1, 0.7))/(1:2000)),
type="l", lty=1, col=1,
xlab="Number of Flips", ylab="Ratio of Head")
```



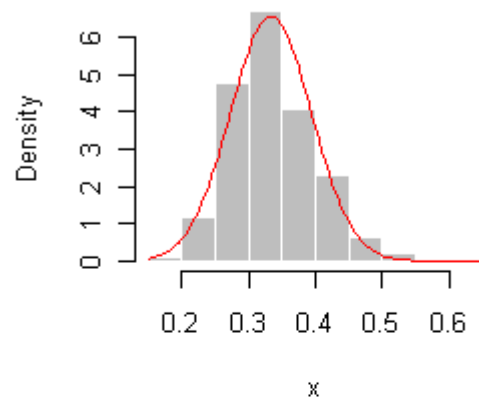
範例：利用生成 1000 組指數分配的樣本平均數驗證中央極限定理。

```
lambda <- 3
size <- c(10, 30, 50, 100)
R <- 1000
par(mfrow=c(2,2))
for (n in size){
  mu <- 1/lambda
  sigma <- 1/(lambda*sqrt(n))
  hist(replicate(R, mean(rexp(n, lambda))),
  freq=FALSE, ylim=c(0, dnorm(mu, mu, sigma)),
  border="white", col="gray",
  xlab="x", main=paste("Sample Size:", n))
  curve(dnorm(x, mu, sigma), add=TRUE, col=2)
}
par(mfrow=c(1,1))
```

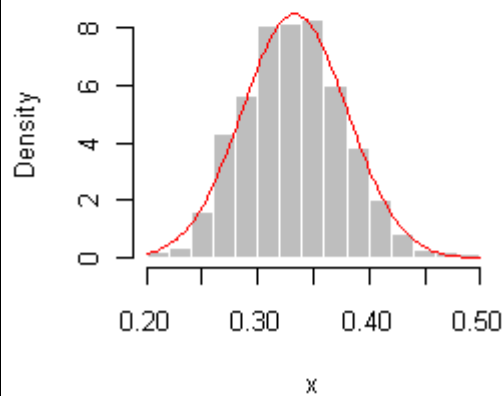
Sample Size: 10



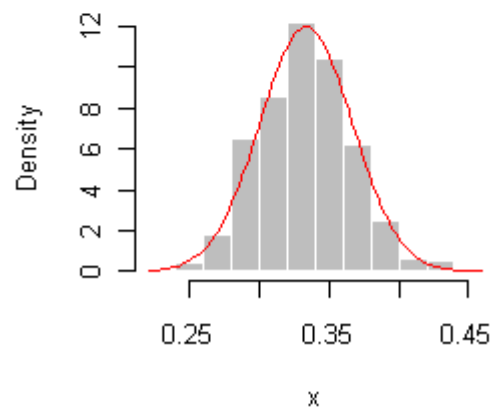
Sample Size: 30



Sample Size: 50



Sample Size: 100



畫一個樹葉

```
#####The Barnsley fern fractal#####
# similar idea to the Sierpinski gasket
#Define the random transformation:
TT=function(xn){
i = sample(c(1,2,3,4),1,replace=TRUE,prob=c(0.01,0.85,0.07,0.07))
  #T1
  if (i==1) TT.r=matrix(c(0,0,0,0.16),2,2)%*%matrix(xn,2,1)
  #T2
  if (i==2) TT.r=matrix(c(0.85,-0.04,0.04,0.85),2,2)%*%matrix(xn,2,1)+
    matrix(c(0,1.6),2,1)
  #T3
  if (i==3) TT.r=matrix(c(0.2,0.23,-0.26,0.22),2,2)%*%matrix(xn,2,1)+
    matrix(c(0,1.6),2,1)
  #T4
  if (i==4) TT.r=matrix(c(-0.15,0.26,0.28,0.24),2,2)%*%matrix(xn,2,1)+
    matrix(c(0,0.44),2,1)
xn1 =TT.r
xn1
}
#Choose an initial point:
x0 = c(1,1)
#The number of steps of the random walk:
N = 10^5
#Initialize an N-by-2 matrix. The rows of X
#will be the N points of the random walk.
X = matrix(0,N,2)
#The first row is the initial point:
X[1,] = x0
#The random walk is given by
for (i in 2:N){
  xn = X[i-1,]
  X[i,] = TT(xn)
}
Qx = X[,1]
Qy = X[,2]
plot(Qx,Qy,cex=0.1,asp=1,col='limegreen',axes=FALSE,xlab=' ',ylab=' ')

mtext('Happy holidays!',col='red',cex=2)
```

```
#####
```

```
#####
```

or, use the following code:

```
N = 10^5
```

```
Xn = numeric(N)
```

```
Yn = numeric(N)
```

```
x=1
```

```
y=1
```

```
Xn[1] = x
```

```
Yn[1] = y
```

```
c.matrix = vector("list",4)
```

```
c.matrix[[1]] = matrix(c(0,0,0,0.16),2,2)
```

```
c.matrix[[2]] = matrix(c(0.85,-0.04,0.04,0.85),2,2)
```

```
c.matrix[[3]] = matrix(c(0.20,0.23,-0.26,0.22),2,2)
```

```
c.matrix[[4]] = matrix(c(-0.15,0.26,0.28,0.24),2,2)
```

```
a.vector = vector("list",4)
```

```
a.vector[[1]] = matrix(c(0,0),2,1)
```

```
a.vector[[2]] = matrix(c(0,1.6),2,1)
```

```
a.vector[[3]] = matrix(c(0,1.6),2,1)
```

```
a.vector[[4]] = matrix(c(0,0.44),2,1)
```

```
for (i in 1:(N-1)){
```

```
  a = sample(c(1:4), 1, replace=TRUE, c(0.01,0.85,0.07,0.07))
```

```
  xy = c.matrix[[a]]%*%c(Xn[i],Yn[i]) + a.vector[[a]]
```

```
  Xn[i+1] = xy[1]
```

```
  Yn[i+1] = xy[2]
```

```
}
```

```
plot(Xn, Yn,cex=0.1,asp=1,col='limegreen',axes=FALSE,xlab=' ',ylab=' ')
```

參考書籍：

1. Heiberger R.M. and Holland, B. (2004) Statistical Analysis and Data Display- An Intermediate Course with Examples in S-Plus, R and SAS, Springer.
2. 陳景祥 (2011) R 軟體-應用統計方法，東華書局。
3. R, and SPSS manual
<http://www3.stat.sinica.edu.tw/stat2005w/download/R.pdf>
<http://genetics.agrsci.dk/~sorenh/misc/Rdocs/RFewHours.pdf>
<http://cran.r-project.org/doc/contrib/Verzani-SimpleR.pdf>
Or download from <http://cran.r-project.org> Contributed Documentation.
4. 路繼先(2004) 統計程式設計講義，成大圖書部。